UNIVERSITY OF MALTA

Faculty of Engineering

**MEC4001 Biomechanics**

# Lab Report – Gait Analysis

by

Eric Borg Saywell (220297M)

Bachelor of Engineering (Hons.) of the University of Malta

20/12/2018

# Contents

## Question 1

### The Gait Cycle

Gait analysis is the systematic study of human locomotion, using the eye and brain of experienced observers, complimented by specific tools for measuring body movements, body mechanics and muscle activity. A complete bipedal gait cycle is defined as a sequence of events of movements during locomotion in a particular time period during which one-foot contacts the ground, to when ground contact of the same foot occurs again resulting in forward propulsion of the centre of gravity of the subject's body. The opposite foot is involved through an identical sequence of events, but displaced in time by half a cycle. A single gait cycle commonly referred to as a stride, consists of two major phases as seen in Error! Reference source not found. below:

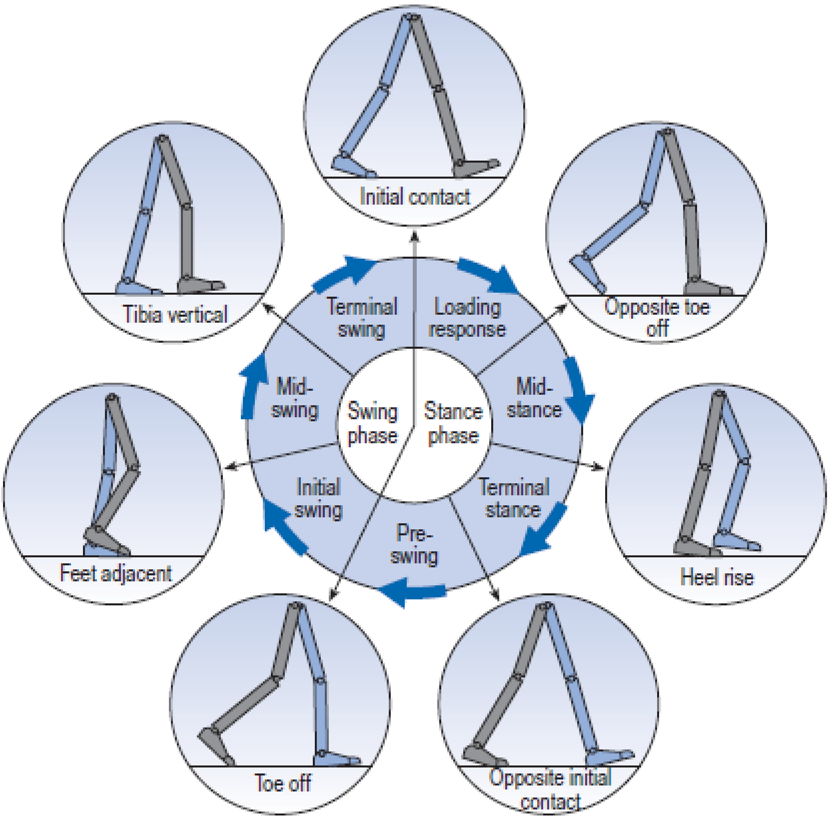



*Figure 1- Leg Positions during a gait cycle. [1]*

Each phase is further decomposed into events as follows [2] :

**<u>The Stance Phase</u> -** The referenced foot remains in contact with the ground beginning with foot strike and ending with toe-off, lasting for about 62% of the whole cycle;

1) Loading Response: 0-12%

Initial contact with the ground occurs with the heel strike where weight is transferred towards the supporting referenced leg in order to maintain speed while balancing the body's centre of gravity by absorbing energy through a "suppressive" action via the lower limb musculature.

2) Mid-Stance: 12-34%

During this phase, toe off of the opposite leg occurs where full charge of the body is supported by the knee and ankle joints of the referenced ground contacting foot allowing the body forward over the foot. This phase corresponds to the first half of the single-foot support and ends when the body's centre of gravity is positioned directly vertical of the supporting foot. The ground contacting foot anchors the body the floor such that it becomes the pivot of the supporting leg.

3) Terminal Stance: 34-50%

This phase corresponds to the second half of the single-foot support where a heel off occurs on the ground contacting foot as it begins to rise to make the next step. The body starts to shift forward of the body's centre of gravity of the supporting leg and swing out to the contact of the opposite foot with the ground to propel the body forward. This phase ends just before toe of the referenced leg loses contact with the ground.

4) Pre-swing: 50-62%

This phase brings an end to the stance phase as the body weight is transferred during initial toe off from the ground contacting leg to the swing leg as it is propelled forward to initiate contact on the opposite leg via the heel strike. Its role is to create an impulse on the forefoot of the trailing leg on the ground in order to push the leg forward.

**The Swing Phase;** during which the referenced foot is not in contact with the ground beginning with toe-off and ending with foot strike, lasting for the final 38% of the cycle;

5) Initial swing: 62-75%

This phase corresponds to the first third of the swing phase. It begins with the referenced leg being lifted from the floor as contact is lost completely from toe-off as work is done to accelerate the leg forward. At this point the ground contacting leg is behind the centre of gravity of the whole body. This phase ends when the swinging foot is opposite and almost parallel to the stance foot.

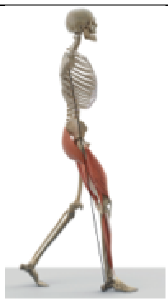
6) Mid Swing: 75-90%

This phase begins when the swinging referenced foot is opposite the stance foot and ends when the tibia of the swinging leg starts to flex and is vertical just before heel off of the referenced leg occurs.

7) Terminal Swing: 90-100%

This phase corresponds to the third tier of the swing phase. The swinging referenced leg is in maximum flexion in mid-air just before initial contact of the next heel strike with the ground occurs for yet another gait cycle.

*Figure 2 - The Phases of the Human Gait Cycle [3]*

| Gait Phases | | Muscle Activity | Hip Joint | Knee Joint | Ankle Joint |
|---|---|---|---|---|---|
| Initial Contact |  | Quadriceps Femoris<br>Anterior Tibialis<br>Gluteus Medius<br>Gluteus Maximus<br>Semitendinosus<br>Semimembranosus<br>Biceps Femoris | 20 ° Flexion | 0° – 5° flexion | 0° |
| Loading Response |  | Quadriceps Femoris<br>Anterior Tibialis<br>Gluteus Medius<br>Gluteus Maximus<br>Adductor Magnus<br>Tensor Fascia Latae<br>Tibialis Posterior<br>Peroneus Longus | 20 ° Flexion | 20° flexion | 5° – 10° plantar Flexion° |
| Mid Stance |  | Gastrocnemius<br>Soleus | 0 ° Flexion | 0° – 5° flexion | 5° dorsal flexion |
| Terminal Stance |  | Gastrocnemius<br>Soleus<br>Flexor Digitorum Longus<br>Flexor Hallucis Longus<br>Tibialis Posterior<br>Peroneus Longus<br>Peroneus Brevis | -20 ° Hyperextension | 0° – 5° flexion | 10° dorsal flexion |
| Pre-Swing |  | Gastrocnemius<br>Soleus<br>Rectus Femoris<br>Adductor Longus | -10 ° Hyperextension | 40° flexion | 15° plantar Flexion° |
| Initial Swing |  | Extensor Hallucis Longus<br>Flexor Hallucis Longus<br>Sartorius<br>Iliacus<br>Tibialis Anterior | 15 ° Flexion | 60° – 70° flexion | 5° plantar Flexion° |

| | | | | | |
|---|---|---|---|---|---|
| Mid Swing |  | Semimembranosus<br>Semitendinosus<br>Biceps Femoris<br>Anterior Tibialis | 25 ° Flexion | 25°<br>flexion | 0° |
| Terminal Swing |  | Quadriceps Femoris<br>Semitendinosus<br>Semimembranosus<br>Biceps Femoris<br>Anterior Tibialis | 20 ° Flexion | 20°<br>flexion | 0° |

## Exporting Data from Gait Analysis

A software called MOKKA, specifically used for analysing kinetic and kinematic motion, was used to transform the data obtained from the test subject during the lab session into a virtual gait cycle. After importing the data into the software, the best possible gait cycle was chosen from the 3 trials performed. The left foot was chosen as the reference, with the gait cycle being initiated upon initial contact of the left heel strike with the ground and ending with contact of another heel strike of the same foot. Since the subject's stride was too long for the pressure plates to record reaction forces against the ground, the succeeding heel strike was derived from the graph. A horizontal line was projected from the z-axis at its respective initial position at frame 336 in order to derive the end of the cycle at the same position which was found to be at frame 449. Thus, in this case, the complete gait cycle was identified and cropped to between frames 336 and 449 leaving only the major events of interest for the cycle as depicted by **Figure 3.**

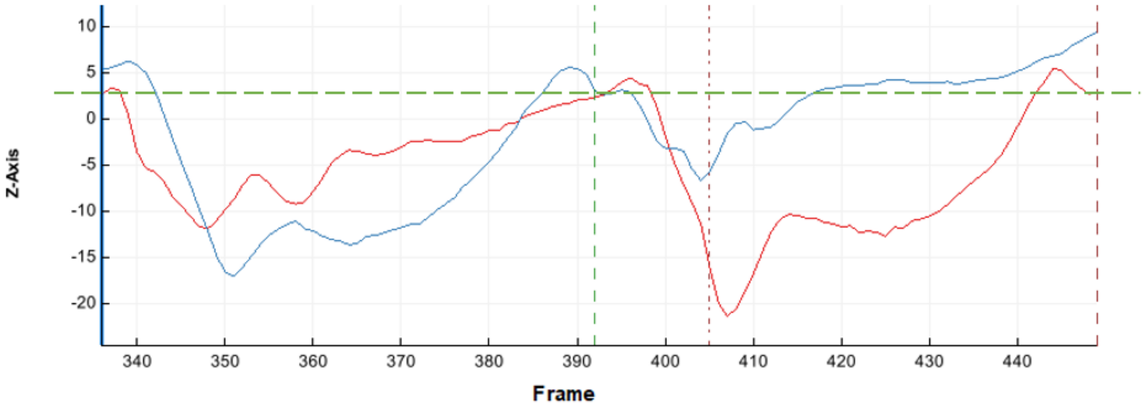
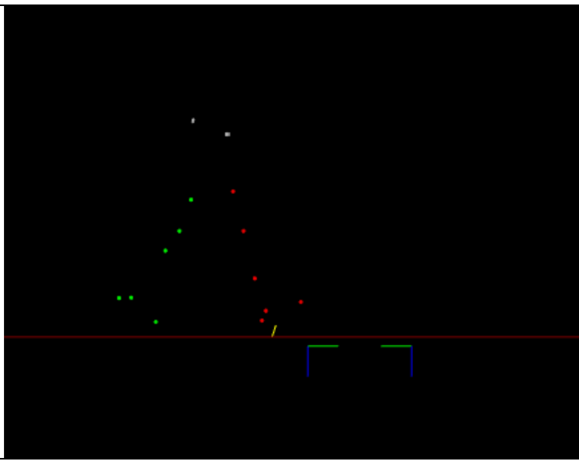
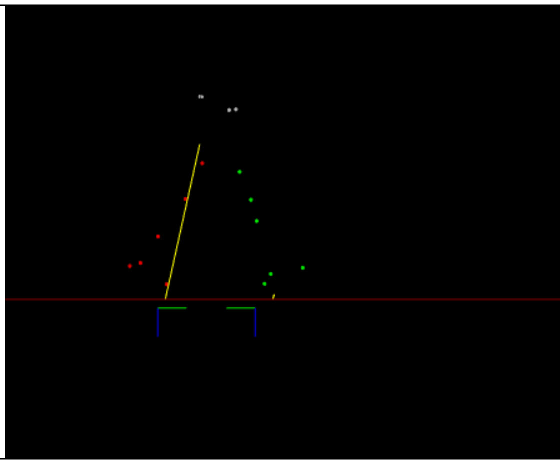


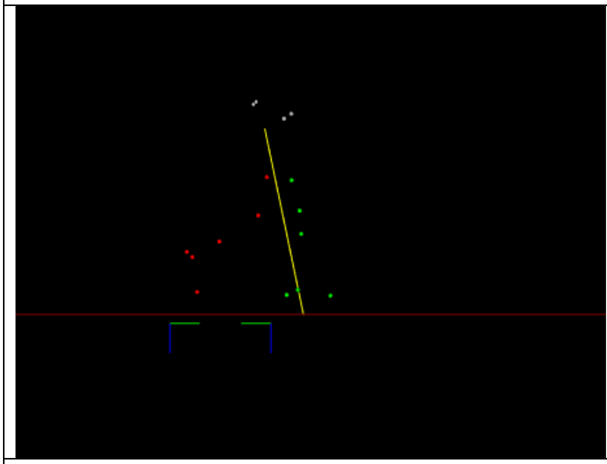

*Figure 3 - Complete Gait Cycle obtained during frames 336 to 449 with the red and blue graphs indicating the left and right ankles respectively.*



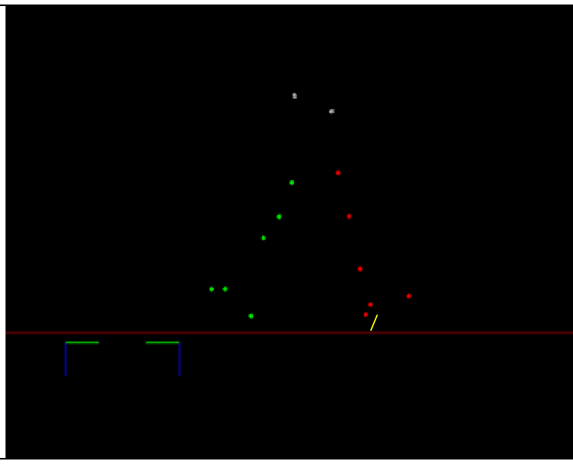
Initial Contact of Left Heel Strike at Frame 336



Right Heel Strike at Frame 392



Left Toe off at Frame 405



Left Heel Strike completing the Gait Cycle at Frame 449

*Figure 4 – Gait Cycle Major Events of Test Subject Walking on Pressure Plates*

## Pedotti Diagram

Error! Reference source not found. depicts the Pedotti diagram showing ground reaction vectors (GRV) and is made up of successive representations, at uniform time intervals, of the magnitude, direction and point of application of the ground reaction force vectors. A plot of the GRV through the gait cycle is called a butterfly diagram due to the pattern produced as the vectors move across the diagram from left to right.

On initial contact during heel off, it can be seen that the GRV is tilted backwards depicting braking forces as a result of the forces produced opposite of the direction of motion. Similarly, the forward tilted GRV during toe off depicts forward propulsion. The vector swings round from backwards at heel-strike, vertically during the middle of stance, then forwards at toe-off. The vector height depicting the magnitude changes continuously; being very small at heel-strike, rising to a maximum just before the ankle is flat against the ground, falls slightly during the middle of stance, rises again and then falls to zero during toe-off when contact is lost. [4]

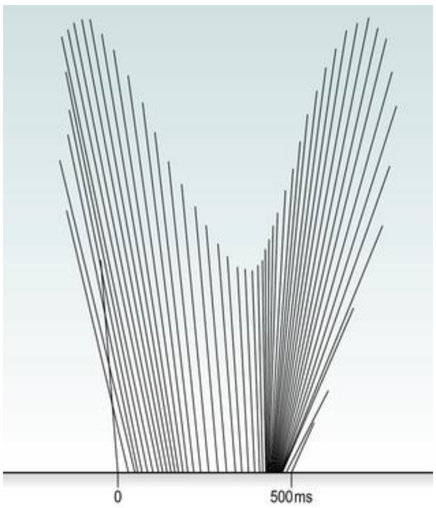
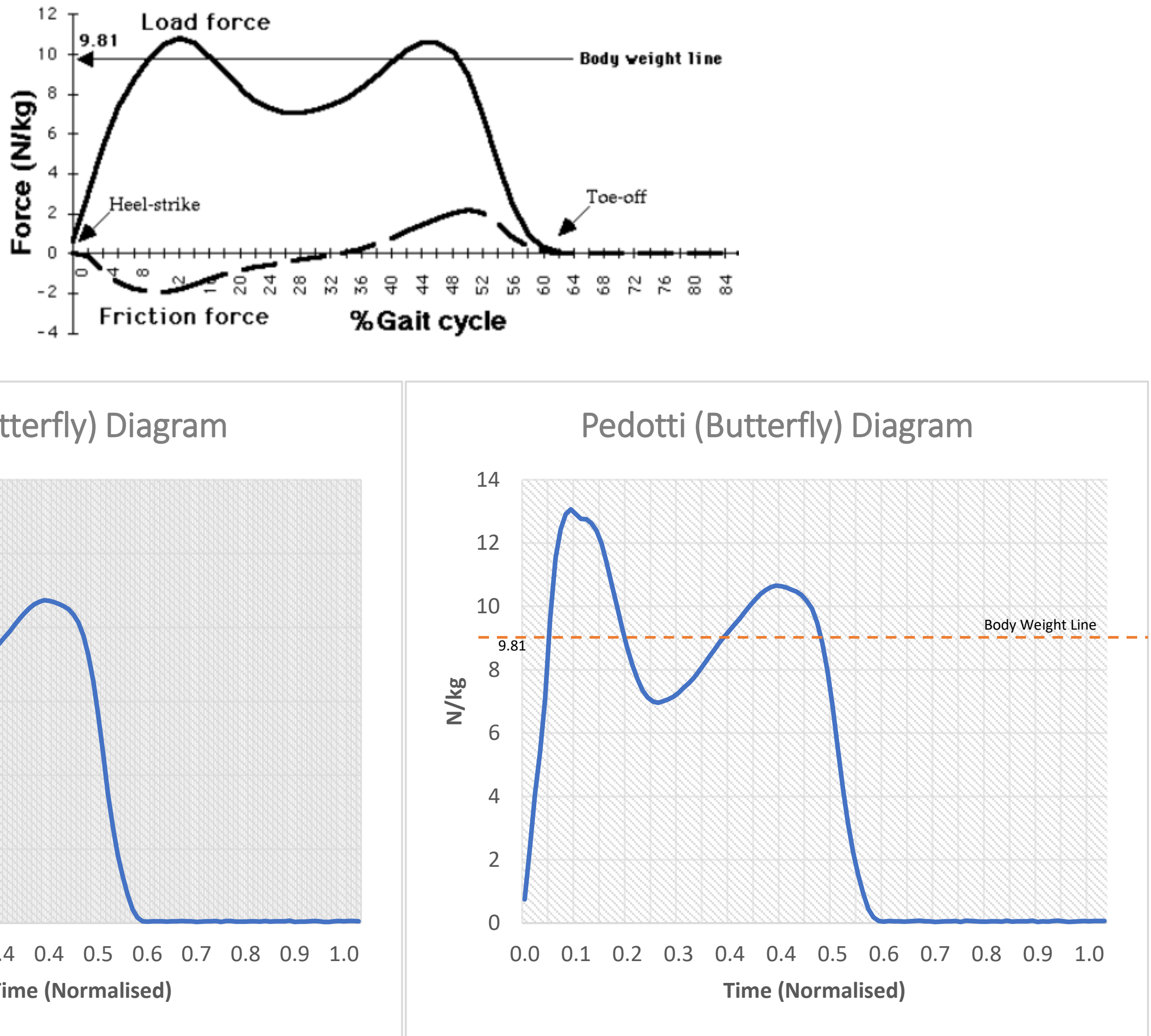



*Figure 5 - Pedotti Diagram depicting Ground Reaction Vectors [4]*




***Top:*** *Theoretical Pedotti diagram depicting Ground Reaction Vectors for body mass normalisation in Normal Gait. [5]*
*Figure 6 - **Left**: Pedotti Diagram depicting Force vs Normalised Time.*
***Right**: Pedotti Diagram depicting vertical oscillations with respect to the test subject's centre of gravity during gait analysis.*

**Figure 6** compares the data extracted from the gait analysis experiment to the theoretical Pedrotti diagram. The bottom left graph depicts the Force produced on the pressure plates with respect to normalised time, while the bottom right graph scales the forces according to body mass normalisation to facilitate interpretation for different types of body weights. When comparing the body normalised Pedotti diagram to the theoretical, it can be seen that the test subject has a much higher degree of vertical oscillations than the typical normal person since the peaks and troughs of the diagram oscillate significantly about the body weight reference line. This shows a great inefficiency in the test subject's gait lacking forward propulsion due to a 'bouncy' stride.

# Question 2

## Gait Analysis Lab Setup

The process of Three-Dimensional Gait Analysis (3DGA) involves the use of various special apparatuses specifically attributed towards capturing motion data to the minutest of details. Any typical gait lab is set up typically as shown in **Figure 7**. As a result of the complexity required to appreciate the amalgamation of gait kinematics, kinetics and muscle activity, a technical description of the instrumentation is provided.

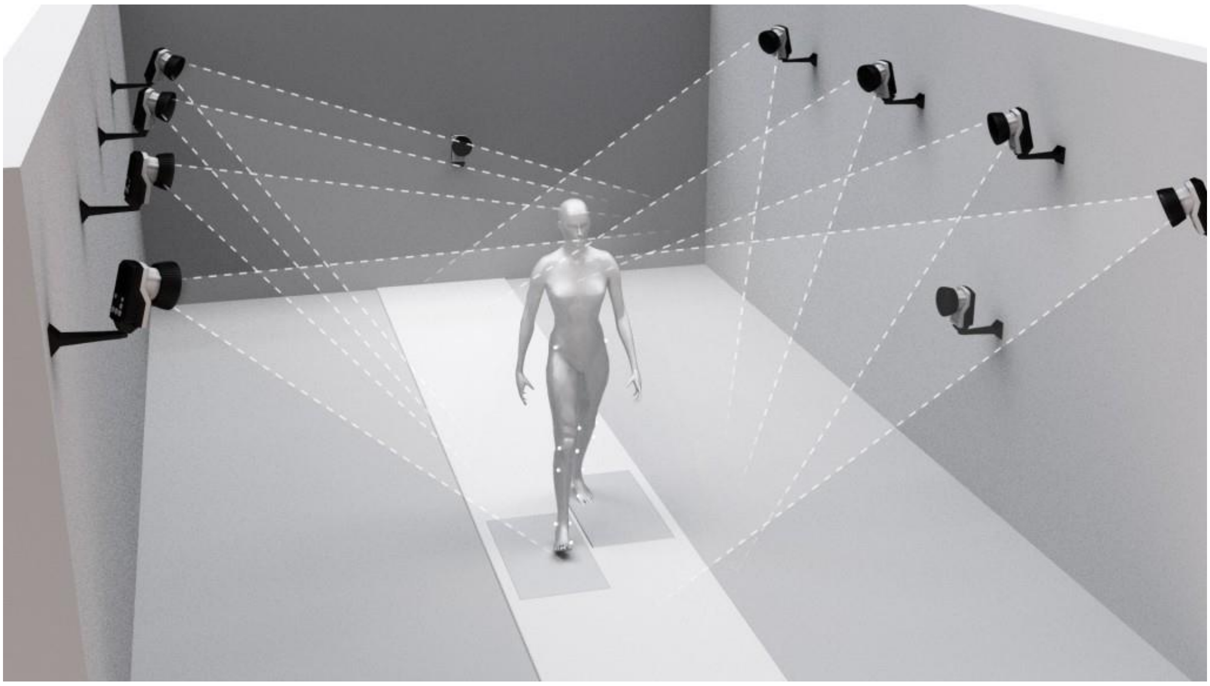


*Figure 7 - Diagram representing the gait analysis lab setup [14]*

## Technical Description of Instrumentation: [4]

**<u>Retro-Reflective 14mm diameter Markers:</u>**

Sixteen infra-red reflecting markers were mounted onto the test subject's joints at predetermined standard locations to capture data on body segment movement during gait. These were programmed to flash on and off in sequence so that only one is illuminated at any instant of time. This prevented interference from each other and eliminated the need for a 'tracking' procedure to determine which one is which. By the use of these active markers in the form of infra-red emitting LEDs, special cameras located and measured the positions of the markers.

**<u>Vicon MX-T10 and MX-T100s Cameras:</u>**

A typical gait lab arrangement usually comprises of 8-12 cameras to cover a capture volume of at least 4m × 1.5m × 2 m (length × width × height). Eight Infrared sensitive cameras were specifically positioned in the gait lab, all pointing towards the capture volume. Infra-red radiation emitted from the markers was detected and processed to capture the test subject's movement from every possible angle to build the 3D model. However, these cameras record not only infra-red light emitted from the LEDs, but any exterior light falling on it. Stray ambient light was fairly easy to eliminate by covering windows and infrared emitting electronic devices were kept hidden at all costs. Possible reflections caused by the markers themselves are usually difficult both to detect and to eliminate but the VICON Nexus motion analysis software is encompassed with a function to adapt to unknown infra-red markings.

**Motion Analysis Software: Vicon Nexus ver. 2.5**

The motion analysis software used to process the captured data was Vicon Nexus ver. 2.5. The graphical user interface allowed monitoring and manipulation of the motion data recorded, as well as other data obtained from other instrumentation from within the motion lab.

**Calibration Wand: VICON Active Wand:**

The Vicon Active Wand V2 is a dynamic calibration device which uses optical markers to synchronize the markers to the infrared cameras by calibrating the brightness of the LEDs to adapt the markers to the specific working environment. Moreover, by moving the wand accordingly within the capture volume at different angles and positions made sure that all infrared markers were distinguished while forming a thorough visualisation of the capture volume.

**Force-Plates: AMTI Force Plates**

Two AMTI force plates were situated at the centre of the capture volume walk way. Within the platform, a number of piezo-electric transducers measure tiny displacements along the upper surface of the force plate via electrical surges. The magnitude of the force is then computed and converted equivalently to the electrical charges by the plates in all three axes, visualising 3D Vectorial forces for the Ground Reaction Forces (GRF).

**Surface EMG: Zero Wire sEMG**

Surface electromyography corresponds muscular activity to voltages generated within the muscular tissues. During the gait analysis, silver chloride electrodes were attached in a way such that they were parallel to their respective muscle fibres. Before attachment, it was made sure that the surface was cleaned from excess dry dead skin and shaved from hair, that could result in erroneous measurements due to an increased resistance off the skin surface. Thus, the charge sent by the motoneurons from the brain to contract the muscle tissue is measured via the electrodes. [6]

**Dynamometer: Mark-10 MesurGauge Dynamometer**

A dynamometer was used to relate the electric potential captured by the EMGs, to the actual forces produced by the muscle fibre. The Mark-10 MesurGauge Dynamometer was used to correlate the obtained data graphically and numerically.

**Miscellaneous**

Anthropometric data for limbs and other body lengths were measured using a measuring tape and an anthropometer.

A stopwatch was used to measure the time of the dynamometer readings in order to synchronize the results with the software.

A calculator was used for simple Newton-to-Kilogram conversions such as mass readings off the force plates as well as other calculations.

# Question 3

## Experiment Methodology

The gait analysis was performed in the Faculty of Engineering Biomechanics Lab.

Before the experiment was initiated, the recommended precautions were taken by closing all windows and doors to ensure that little to no infrared radiation from external sources could access the lab. This was done to minimize and eliminate all possible interferences with the infra-red cameras. Moreover, all reflective material on the test subject or capture volume was covered in masking tape to remove reflective media that could tamper with the cameras. Soon after, the VICON Nexus Software and Giganet Servers were activated together with all sEMGs and Force Plates required for the test.

When setting up the Vicon Nexus Software, it was made sure that a good communication was active between all equipment prepared for the test to have an efficient synchronization. The 'GO LIVE' function was activated from the software and all cameras, force plates and sEMGs were marked as green indicating all systems were online and synchronized.

From within the software, Data Management was accessed to create the patient and session data together with the Patient Classification. The gait test subject was added by clicking on the yellow circle and was labelled anonymously. A new session was then added and was named according to the date of the test day for traceability purposes.

From the Vicon Nexus software, all cameras were selected from the System Tab in order to start calibration. From Camera View mode, the capture volume was visualised and was used to recognise any exposed reflective material or undesirable luminous sources. Regardless of covering all apparent reflections, unwanted light inclusions were still present which were removed by camera masking. By navigating to the Tools toolbar, the "Mask Cameras" setting was selected and initiated by clicking on the Start button. Upon making sure that all visible white dots on the camera views were covered by blue pixels, the masking process was stopped.

Moreover, the "Show Advanced" button was selected and "Active Wand v2" was selected under the Wand menu. The "Full Calibration" setting was chosen under the calibration type. The VICON Active Wand was turned on and set within the capture volume. After starting the calibration using the "Start" command, the wand was spun around the capture volume. Calibration is stopped automatically when approximately 1000 wand counts were detected by each of the cameras. The image error was then checked from the software, ensuring that each error is below 0.9 and marked as green. If camera angles are unsatisfactory, they must never be tampered with as this would result in an obligatory re-calibration of the whole system requiring a technician and overwriting of the system's calibration file.

The global origin was set up by defining a Volume Origin. From the Tools toolbar, the "Camera Tab" was selected and then "Set Volume Origin" and "Show Advanced" were selected. The Active Wand was then positioned on the gait pathway of the capture volume in between the force plates and was levelled using the inbuilt water leveller of the wand, making sure the level is marked in the balanced zone. Subsequently, it was made sure the wand was positioned just enough to touch the edges of the force plates. Under “set Volume Origin”, the “Start” command and “Set Origin” were clicked on to make sure that the new origin was listed under the 3D perspective view and confirmed to be in the precise position. Finally, the last devices requiring calibration were the force plates. This involved zeroing the Force Plates by pressing the orange buttons found on the Force Plate Amplifiers. Similarly, this was done through the software by navigating under the Device heading in the System Tab and selecting the “Zero Level” option upon right-clicking on each plate.

Right after calibrating the hardware, the test subject also underwent a similar procedure, in order to be calibrated. For new subject registration, the “Subjects” tab was selected under the “Resources” section on the Vicon Nexus software. The “Create a new subject from labelling skeleton” option was chosen and “Plug in Gait” was selected. After labelling the subject, the participant’s anthropometric data was added. The initial measurement was the subject’s body mass which was recorded by zeroing the active force plate and by selecting the GRFz (Ground Reaction Force in Z-Direction) graph from the “Devices” option under the “System” tab. The graph was adjusted accordingly to represent values in Kilograms, by dividing by 9.81 and converting to Newtons. Furthermore, a measuring tape was used to gather anthropometric data on the subject’s height and his leg length by measuring from the hip bone to the medial malleolus of the ankle. Later, the anthropometer was used to measure the distance between the Anterior Superior Iliac Spines (ASIS) as well as the knee and ankle widths between the medial and lateral epicondyles and malleoli respectively. All length dimensions were recorded in millimetres.

After all the required anthropometric data was recorded, the reflective markers were attached to the test subject at various predetermined positions. When all markers were attached, a static calibration was carried out on the participant by posing in a T-shape., with feet shoulders width apart. From the VICON software, upon initiating the image capture using “Subject Capture”, approximately 200 frames were captured, highlighting the markers as grey spheres. Once the capture was stopped after reaching the recommended number of frames, each individual marker was selected and labelled accordingly using the Edit function under the Label menu. All data was then saved.

Once all markers were attached, the surface EMGs and silver-chloride electrodes were also attached to the chosen muscles. The major muscles selected were:

- Gastrocnemius Lateralis (lateral head of the calf muscle).
- Gastrocnemius Medialis (medial head of the calf muscle)
- Anterior Tibialis (lateral to the tibia)
- Vastus Medialis (deepest quadricep muscle)
- Biceps Femoris (Hamstring)

Prior to attaching the electrodes, it was made sure that the test subject was wearing the appropriate clothing in order to have access to the skin above the muscles being investigated. Extra hair was shaved and cleaned using alcohol pads to decrease resistance and achieve more accurate readings from the electrodes. The latter were placed in such a way that they were parallel to the direction of the muscular fibres. The wires and sEMGs were secured to the participant by using masking tape to prevent detachment. Once all sEMGs were attached, the subject was then asked to contract and relax the chosen muscles to confirm that the baseline signal was being detected. Noise signals were checked and it was also made sure that all offsets and possible shifts between contracted and relaxed muscles were zeroed.

Afterwards, the system was ready for the dynamic subject capture. The Record icon was clicked on to start recording the trial. The Trial Type was changed to “Dynamic PiG” and was named as “Walk_001”. As a precaution, the force plates were zeroed again and the “Start” button was pressed to start capturing a dynamic session. The subject was instructed to walk towards the end of the walking path. However, since the test subject’s strides did not concur with the force plates, this process was repeated for numerous times trying out different starting positions, such that a complete gait cycle could be captured whilst also reading contact forces on the force plates.

During a second trial, labelled as “Walk_002”, the test subject succeeded in fitting almost all the gait cycle in the range of the force plates apart from the last heel strike. However, since the test subject was forcing to coincide with the force plates by shortening the gait stride length, given that forcing a short stride would affect the results of the gait cycle, the data was considered satisfactory. The gait was then cropped to the frames of interest containing major events using the “Crop” command. The Gap Filling function was used to fill potential gaps in the motion capture data, accessed under “Edit” in the Tools section. Once the data obtained was clean and satisfactory, all marker trajectories, force-plate data and sEMG data were exported in the ASCII format to create a .c3d file in the parent directory.

The force vs sEMG relationship was studied by connecting a dynamometer to a laptop. One the connection was secured, all sEMGs were removed except for the one attached to the Vastus Medialis. This sEMG was chosen to be investigated as a result of it isolation at the last degrees of knee extension. The test subject was then instructed to sit down on a chair facing the dynamometer setup while connecting his ankle using an ankle strap. The sEMG and dynamometer recordings were synchronized together while using the VICON Nexus and MeasurGauge software respectively for each.  whilst a stopwatch was also started to synchronize results later. The test subject was asked to contract the Vastus Medialis muscle until Maximum Voluntary Contraction (MVC) was reached, while the readings from the software were being recorded simultaneously. Once MVC was reached, the subject relaxed the muscle and the recordings were all stopped. All data obtained from MeasurGauge was extracted to a .csv file while the Vicon Nexus readings were exported to ASCII format.

## Question 4

### Hip Analysis

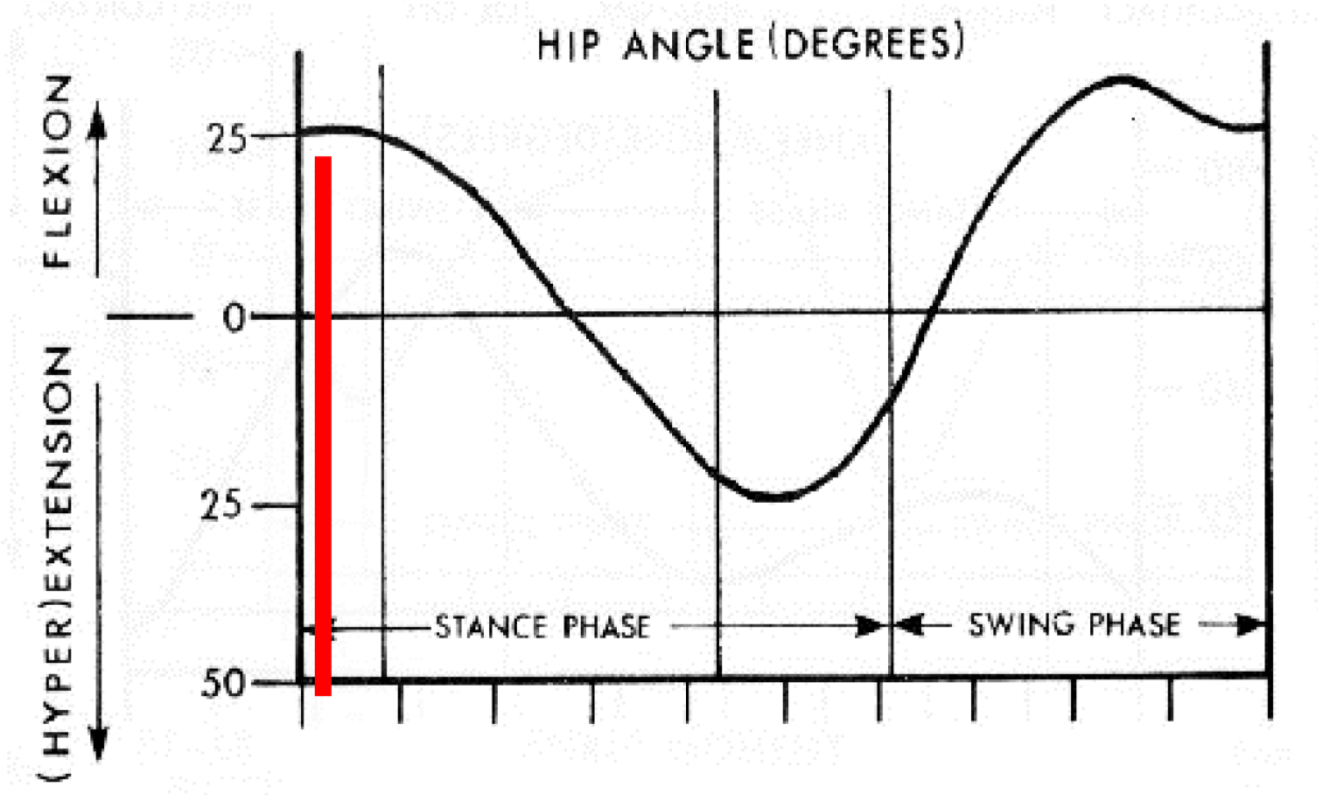



*Figure 8 - Graph of Theoretical Left Hip Angle*

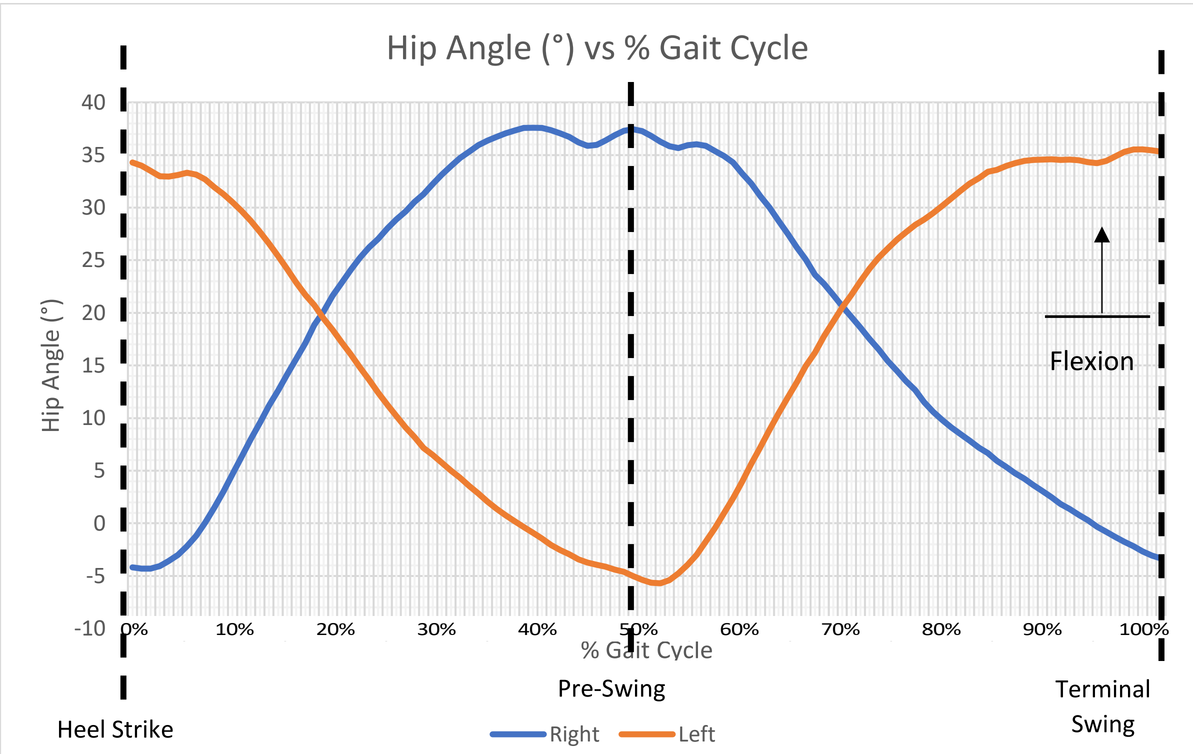



*Figure 9 - Graph of Experimental Hip Angle (°) vs % Gait Cycle*

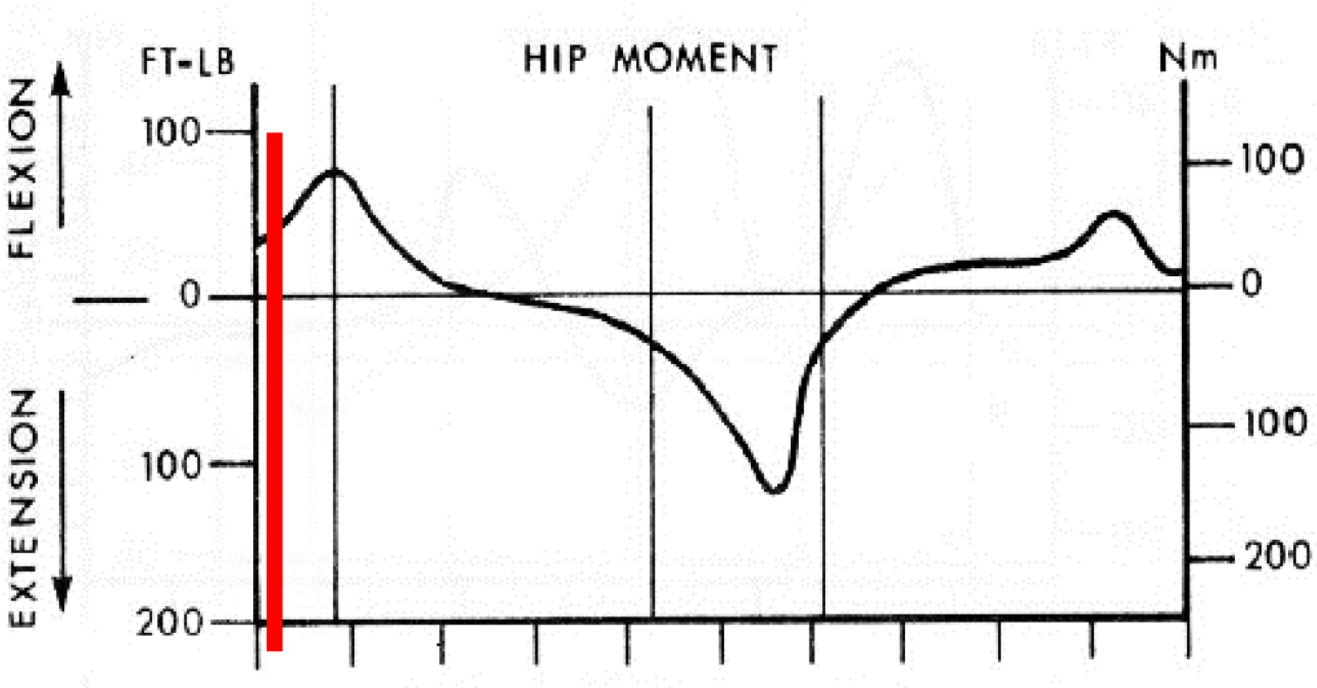



*Figure 10 - Graph of Theoretical Left Hip Moment (Nmm)*

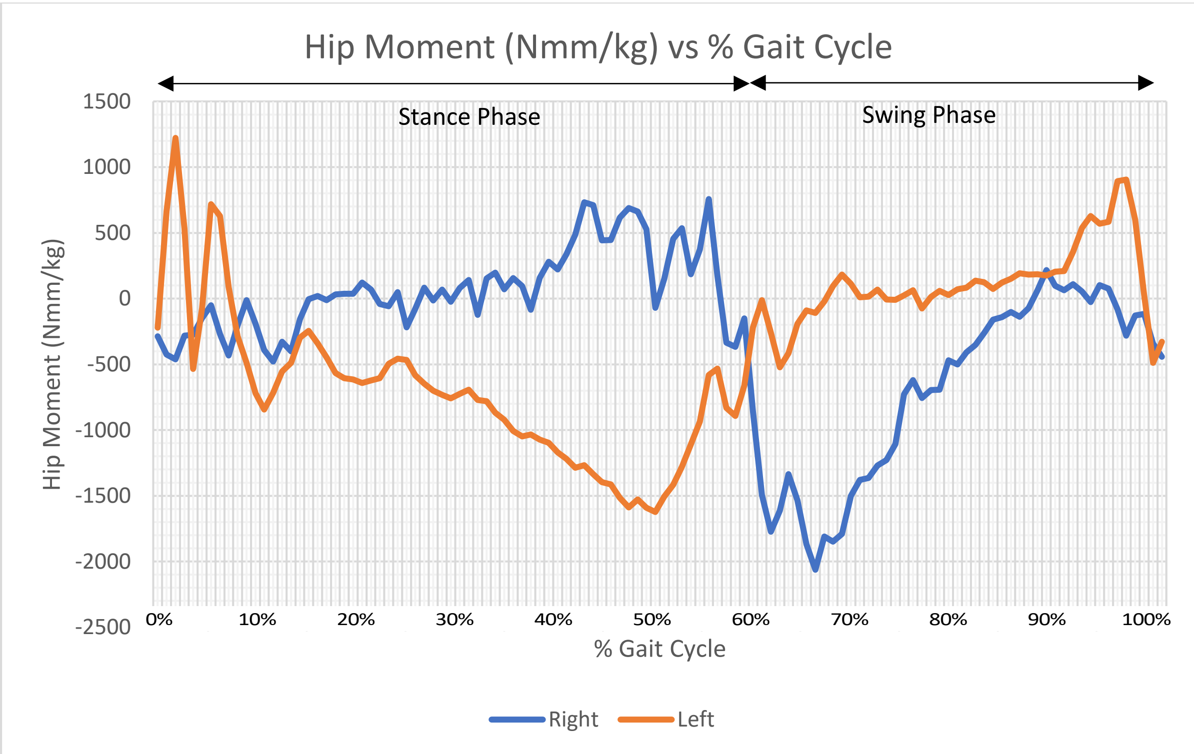



*Figure 11 - Graph of Experimental Hip Moment (Nmm) vs % Gait Cycle*

**Figure 8** and **Figure 9** depict the theoretical and experimental hip angle graphs respectively. At first glance, both graphs of left hip angles look relatively similar, with maximum hip flexion occurring during heel strike and in the terminal swing. Maximum hip extension occurred at the beginning of the swing phase of the opposite leg. It can also be noted that the test subject achieves a higher hip flexion in comparison to the theoretical graph and a lower hip extension showing an average offset of about $10^{o}$ towards flexion. This experimental nonconformity could be as a result of tight hip flexors such as the rectus femoris and tensor fascia Latae, resulting in failure to achieve a higher extension during the pre-swing, resulting in a compensation of the hip flexors to do more work and thus extend less resulting in a shortened stride. Another reason for this experimental anomaly was that during the gait cycle the test subject forced a shorter stride in order to coincide with the force plates, limiting hip extension and thus resulting in a premature hip flexion. [7]

The 'hip' angle may be measured in two different ways; the angle between the vertical and the femur, and the angle between the pelvis and the femur. The latter is the 'true' hip angle and is usually defined so that 0° is when the individual is in the standing position. As can be seen in **Figure 12**, the test subject seems to suffer from an anterior pelvic tilt. This results in a forward flexion of the trunk and pelvis, appearing as a hip flexion when the hip angle is defined with reference to the pelvis, resulting in a tilted pelvis and thus affecting the experimental hip angles by a great degree. Anterior pelvic tilt is also a consequence of tight hip flexors. [7]

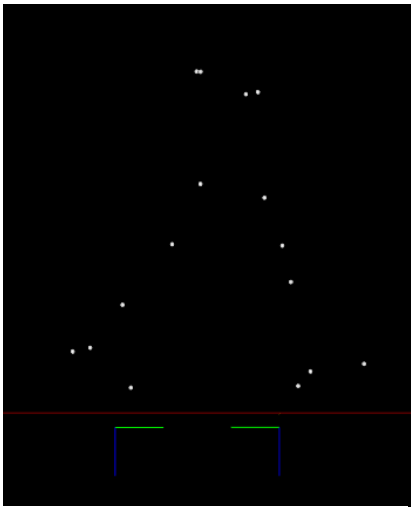


*Figure 12 - Anterior Pelvic Tilt depicted in the pelvic region as a result of the forward rotation of the pelvis.*

**Figure 10** and **Figure 11** depict the theoretical and experimental graphs for the hip moments respectively. On comparing both left hip moment graphs, it can be seen that exactly after the heel strike occurs, hip flexion moments increase to a maximum as a result of eccentric contraction of the posterior chain muscles. This moment immediately changes direction as maximum hip extension is reached and the loading response is underway where the test subject propels himself forward as load is transferred to the quadricep femoris where it continues until the pre-swing phase. The moment again changes direction at toe off as the hip is propelled into the swing phase. This flexion moment continues to increase due to stretching of the hip ligaments and hip flexors such as the iliopsoas. The flexion moment reaches a maximum prior the tibia becomes perpendicular to the ground where maximum hip extension of the gait stride is reached in the terminal swing-phase preparing for the next heel strike. [7]

Knee Analysis

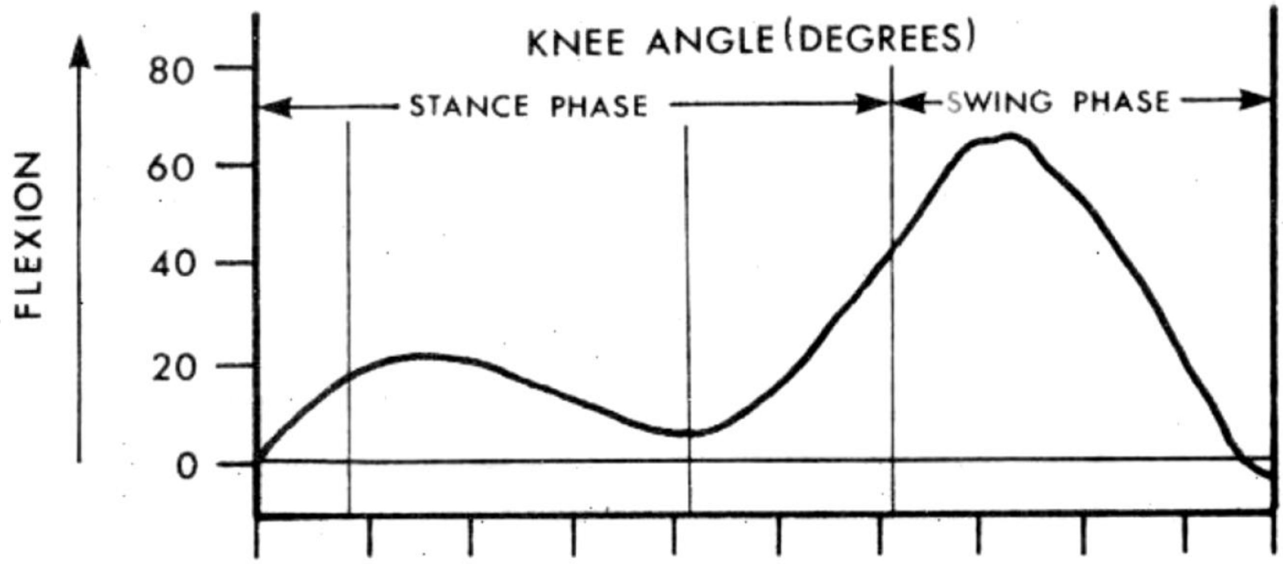



*Figure 13 - Graph of Theoretical Left Knee Angle*

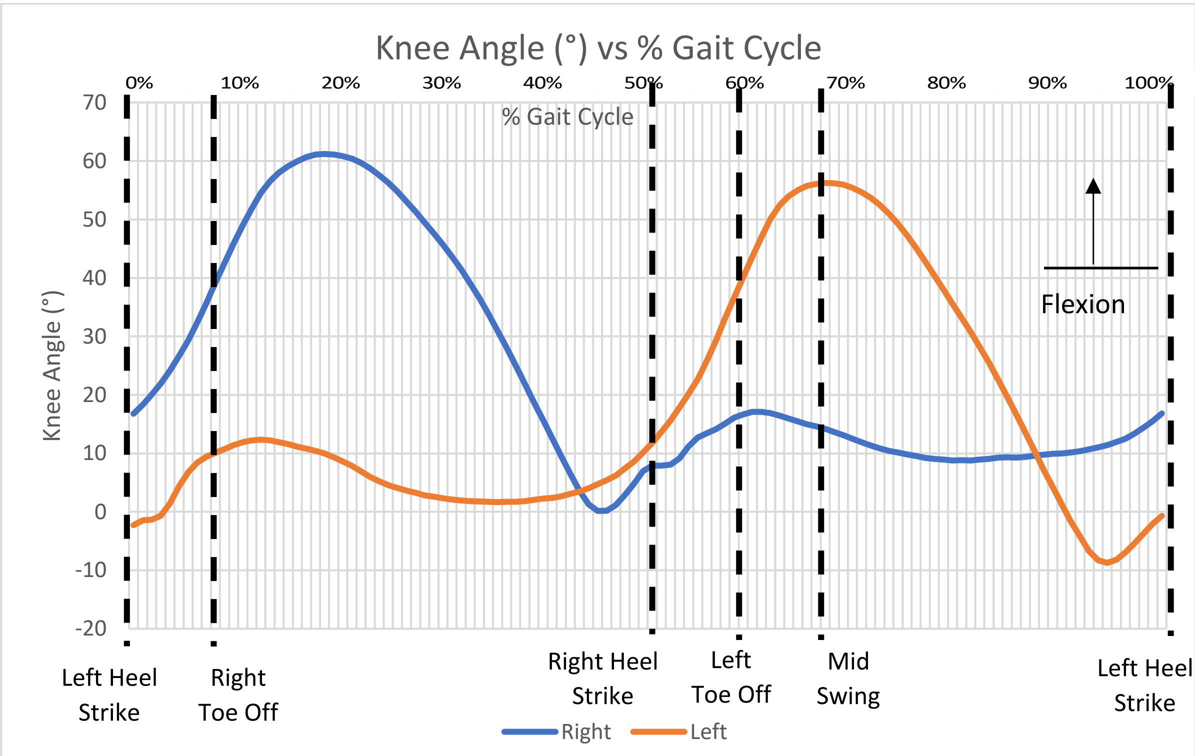



*Figure 14 - Graph of Experimental Knee Angle (°) vs % Gait Cycle*

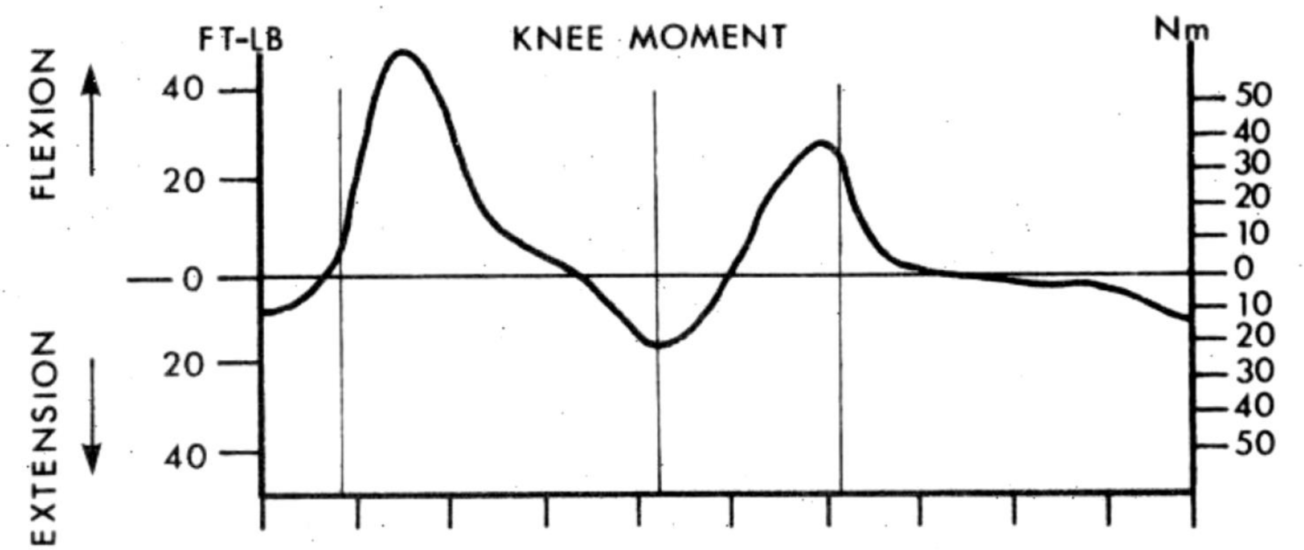



*Figure 15 - Graph of Theoretical Left Knee Moment (Nmm)*

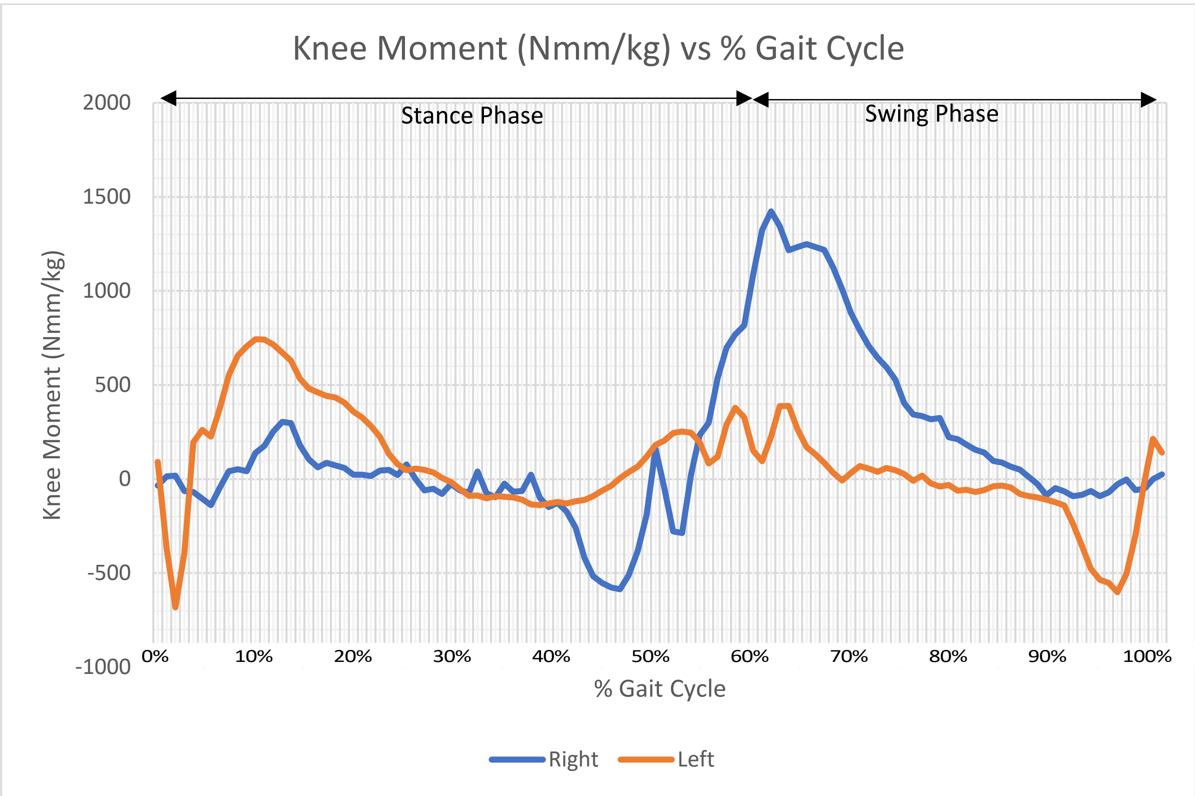



*Figure 16 - Graph of Experimental Knee Moment (Nmm) vs % Gait Cycle*

**Figure 13** and **Figure 14** depict the theoretical and experimental knee angle graphs respectively. At first glance, both graphs of left knee angles look relatively similar except for a slight overall shift downwards towards extension, indicating a negligible amount of hyperextension. The knee angle is defined as the angle between the femur and the tibia and the angle is $0^{o}$ when both are straightly aligned. The extension is most noticeable when the hip is reaching out to perform a heel strike, right before and during the heel strike itself occurring between 97% and 3% of the gait cycle. The overall shape of the graphs conforms well to the theoretical shape depicting 2 knee flexion peaks and 2 extension peaks. At the peak, knee flexion is at 12.32°, occurring just after the loading response, during the early mid-stance. On the theoretical graph, it can be seen that the peak is approximately 20°. However, this is highly dependent on the walking speed, and since the test subject was using a shorter stride as previously explained, the overall shift downwards is vindicated. [7]

Following the first peak, the quadriceps contract concentrically until the heel rise occurs (~36%). At this point, the gastrocnemius is contracted to counter dorsiflexion in the ankle whilst acting as a flexor to the knee. The flexion induced by the gastrocnemius prevents hyperextension and starts flexion up to the second flexion peak. Flexion is accelerated by the movement of the ground reaction vector during the pre-swing phase. Afterwards, flexion is increased due to inertia coming from hip flexion, peaking prior to both tibiae aligning vertically. The second peak occurs at ~67% in the experimental graph, in agreement with the theoretical graph. Knee flexion is stopped slowed down by the eccentric contraction of both the rectus femoris and the hamstrings, leading to full extension at 91% of the gait cycle followed by slight hyperextension leading into the next heel strike. [7]

**Figure 15** and **Figure 16** depict the theoretical and experimental graphs for the knee moments respectively. Overall, the two graphs have similarities but at certain points are distinct due to different characteristics exerted by the test subject during the gait. The first part in each graph are similar, showing a peak in maximum knee extension moments just after the heel strike occurs. In comparison to the theoretical, the experimental knee extension moment is approximately 68Nm when compared to the 15Nm as a result of the hyperextension experienced just after the heel strike. This occurs as a result of the extra tension and work done by the quadriceps and patellar tendons due the slight hyper extension of the knee and also due to the eccentric contraction of the hamstring. During the loading response, a concentric contraction of the hamstrings continues to increase in flexion moment until the opposite toe off where the quadriceps contract eccentrically to counter the flexion moment due to the ground reaction force for forward propulsion. An eccentric contraction of the hip flexors' such as the rectus femoris follows, to counter the knee flexion coming from the hip flexion, resulting in an extension moment which continues until the following heel strike. [7]

A possibility that might explain the large amount of noise and lack of smoothness in the graphs is the fact that the test subject is well endeavoured in physical activity. This results in more worn out joints, ligaments and tendons than the typical person resulting in increased internal moments due to worn out synovial fluid; acting as a lubricant between joints, and thus leading to an increase in friction and internal moments. Moreover, the friction between the tendon and the ligament results in the ligament being exposed to shear loads in addition to normal loads. Additionally, increased shearing forces occur as the tendons attempt to slide through their synovial tunnels creating excess fluctuations in internal moments. [7] [8]

Ankle Analysis

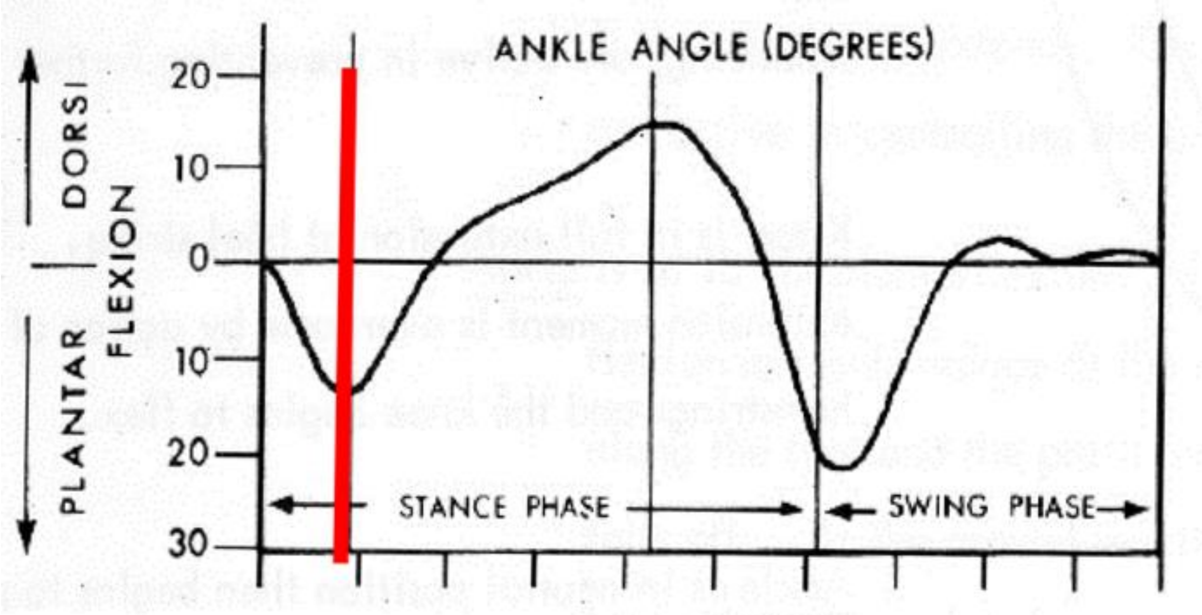



*Figure 17 - Graph of Theoretical Left Ankle Angle*

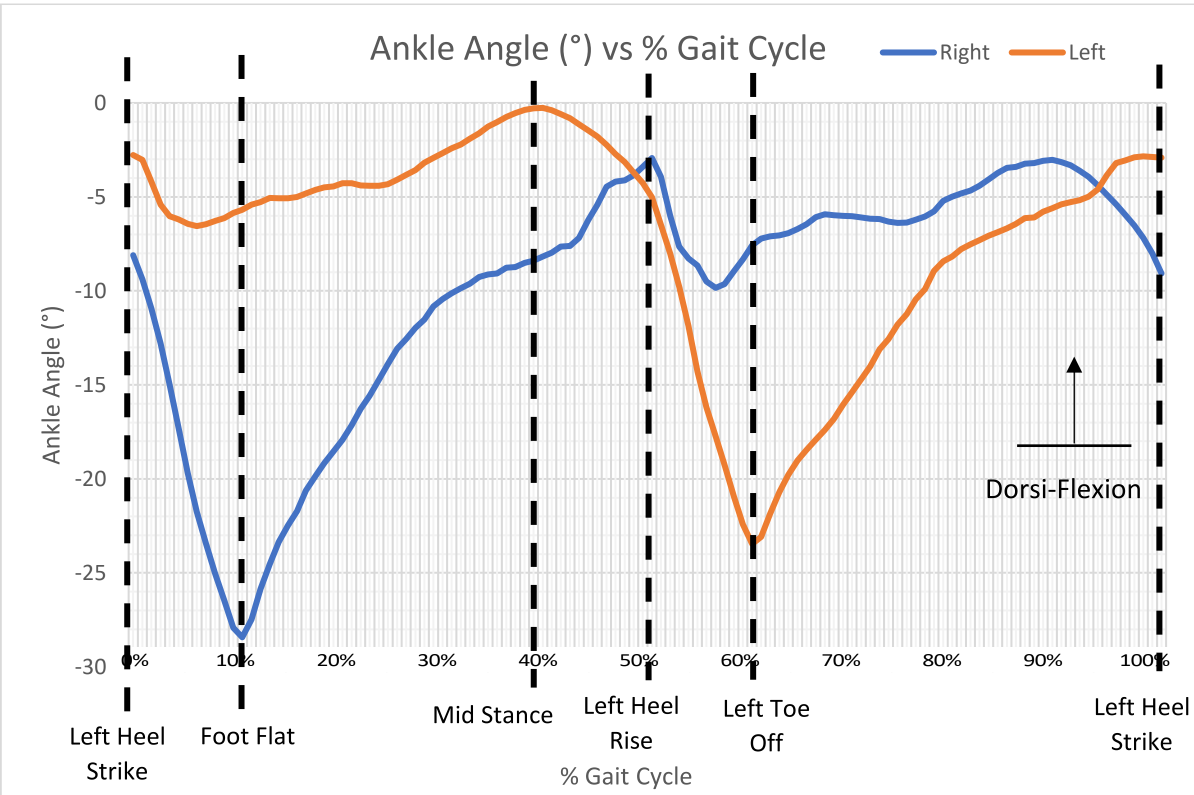



*Figure 18 - Graph of Experimental Ankle Angle (°) vs % Gait Cycle*

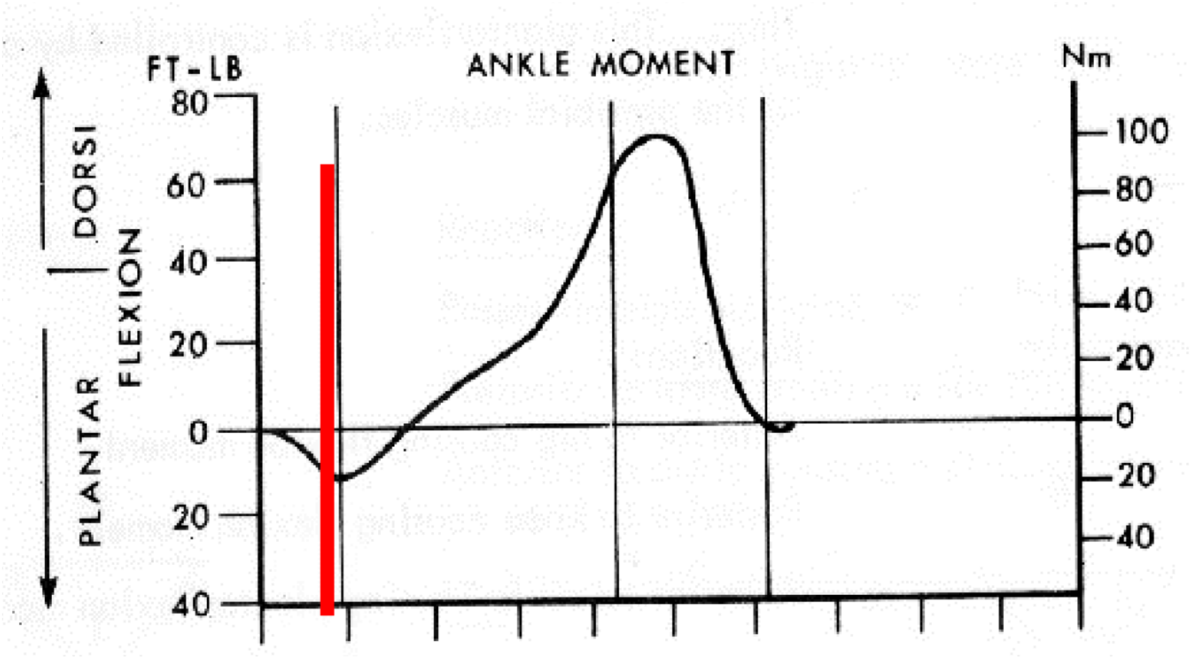



*Figure 19 - Graph of Theoretical Left Ankle Moment (Nmm)*

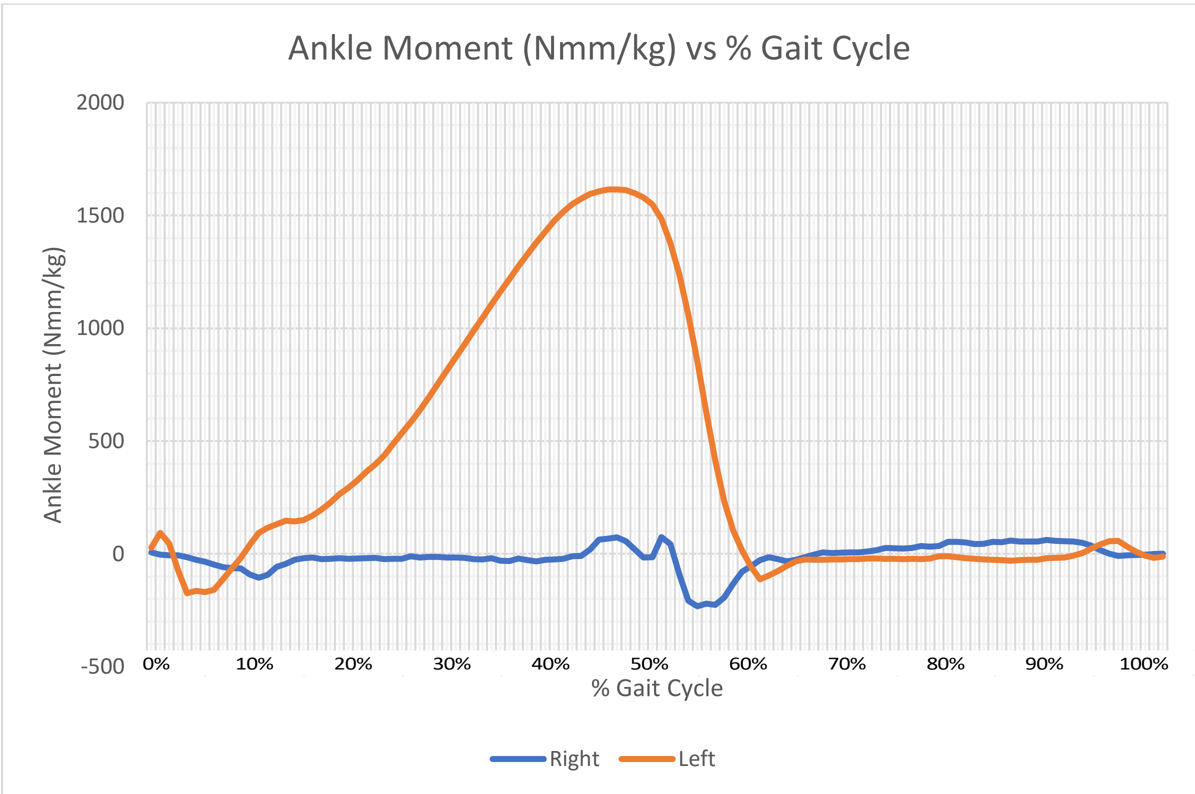



*Figure 20 - Graph of Experimental Ankle Moment (Nmm) vs % Gait Cycle*

**Figure 17** and **Figure 18** depict the theoretical and experimental ankle angle graphs respectively. At first glance, both graphs of left ankle angles look fairly similar. One characteristic that is quite peculiar is the extreme degree of plantar flexion the test subject demonstrates, which results in a great overall downward shift of the graph. The ankle angle is usually defined as the angle between the tibia and an arbitrary line in the foot. Although this angle is normally around 90°, conventionality defines dorsiflexion at an angle of 0° or greater, and plantarflexion being movements in negative angles. Thus, from this statement it can be concluded that theoretically, the test subject never managed to carry out complete dorsiflexion during any part of the gait since all angles were negative. [7]

Right after the heel strike, plantarflexion increases slightly as the foot is flattened due to eccentric contraction of the anterior tibialis. Here, the participant has a lower degree of plantarflexion in comparison to the theoretical. The maximum peak occurred during the mid-stance, with the closest the participant was to achieving ankle dorsiflexion. However, this was still approximately $15^{O}$ shy from the peak of the theoretical graph. During midstance, the left leg, specifically the ankle would be completely weight bearing of the full body weight which usually leads to flexion of the ankle as a result of the loads. However, in this case the latter did not occur, possibly due to a considerably high amount of ankle stiffness the test subject suffers from as a result of tight gastrocnemius muscles as well as the soleus and anterior tibialis. Succeeding the peak in the graph, the ankles experience maximum plantarflexion once again until toe off has been reached. [7]

It is fair to point out that the test subject is well involved in sports, specifically in the sprinting events. Sprinters have a tendency of developing stiff ankles due to forcing cocked dorsiflexed ankles during sprinting since this is greatly encouraged to elevate ground forces by stopping the lower leg abruptly upon impact and avoid amortization of the ankle. [9] The experimental data overall has an ankle range of $24^{o}$ between the extreme angle values while the theoretical graph has a range of $36^{o}$. Thus, the experimental graphical data is significantly flat in comparison to the theoretical, depicting a lack of ankle mobility. Since sprinters tend to keep their ankles cocked during activity, this action is reflected during the gait where the test subject also tends to keep his ankle stiff, as shown in the flatness of the experimental data. On the other hand, the exaggerated plantar flexion during toe off could also be a side effect of the test subject's sporting background, where during triple extension from a block start, a sprinter tends to exaggerate the use of the toe flexor muscle group to achieve maximum forward propulsion from the ankle extension. [7] [10]

Another possibility for the above-mentioned anomalies are the forced shortened strides which were attempted to fitting the whole gait cycle on the force plates resulting in alterations to the test subject's gait. Dorsiflexors such as the anterior tibialis might have not been fully activated due to the shorter steps, supporting the experimental data. Other factors contributing to the lack of dorsiflexion might be anatomical features. The test subject has high-arched feet. Due to this natural anatomical wedge, the foot naturally leans towards plantarflexion resulting in an angle increase between the tibia and the ankle.
Furthermore, given that the participant was wearing athletic shoes with soles, the elevated heel support may have contributed to the excessive plantarflexion. The fact that the ankles are at the base of the body, and certain angles are not reachable due to certain physical constraints of the test subject, might result in compensation and adaptation of higher up

muscles to move differently, thus drastically affecting the test subject's overall gait in comparison to the theoretical graphs. [7]

**Figure 19** and **Figure 20** depict the theoretical and experimental graphs for the ankle moments respectively. Both graphs seem to be identical, showing minimal difference from the theoretical graph. Initially, during heel strike, no moment is present. Just afterwards, the ankle exhibits a small dorsiflexion moment to counter the plantar flexor moment from the ground reaction force. The effects of both moments are demonstrated by the trough in the curve followed by the peak just before mid-stance. From the experimental graph, the peak moment produced was approximately 160Nm in comparison to the 100Nm of the theoretical, which is more than half the work. This illustrates the degree of muscle activity taking place in the calf muscle as a result of the test subject keeping his ankle cocked during the gait. This moment is caused by the extra work being done by the gastrocnemius and soleus as the tibia moves forward, to absorb power off the ground reaction forces until the point of toe-off, where again ankle moment drops to 0 during the swing phase. [7]

## Question 5

The force profile of the Vastus Medialis was obtained during the MVC by using the dynamometer and MeasurGauge software at a sample rate of approximately 50Hz while the sEMG unit was collecting voltage data at a sample rate of 1000 Hz. To synchronize the force and voltage data, the voltage required down sampling. To start off, full wave rectification of the signal was compulsory to change the negative voltages into positive. Filtering of an unrectified EMG signal would have smoothened out the graph with a result close to 0 Volts on either side of the graph due to the EMG's average signal being slightly below 0

The signal envelope would be useless without rectification, thus, by using Microsoft Excel the absolute function was used to rectify the signal. Afterwards, a Butterworth filter add on was imported into the Microsoft Excel Software to be able to rectify the data and process it through an unmodified Butterworth filter with a single, constant cut-frequency. The sheet format was set up as specified by the add in template, with time, data and the output filter put adjacent to each other with the required cut off frequency above the filter column, ranging between 1-20Hz. Afterwards, CTRL-SHIFT-B was used to filter and compute the data in their respective columns. A graph of the filtered data was plotted against time to determine the suitable cut-off frequency. Although trial and error was used, 3Hz was chosen since for the cut-off frequency since the resulting graph was smoother without neglecting data.

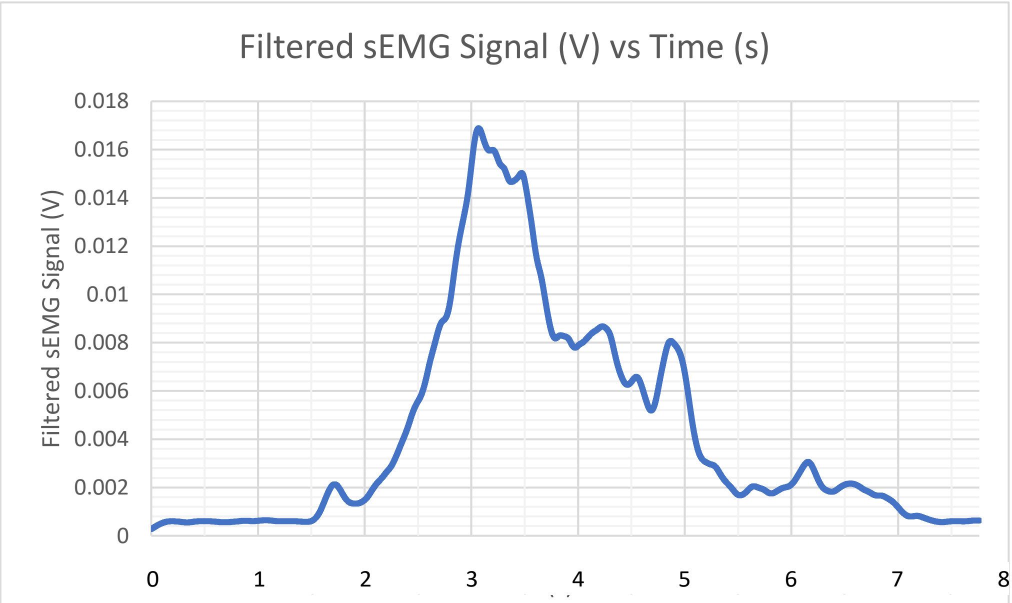



*Figure 21 - Filtered sEMG Signal vs Time*

The dynamometer's time recordings were then imported and used to down-sample the filtered sEMG signal. The LOOKUP function was used to locate and extract the filtered voltages for the corresponding dynamometer force readings to extract a data set based on the dynamometer sample rate after which the filtered signal was normalised. Similar to what was done for the sEMG signal, the data was then imported into the Microsoft Excel spreadsheet to rectify and normalise the force values using the ABS function.

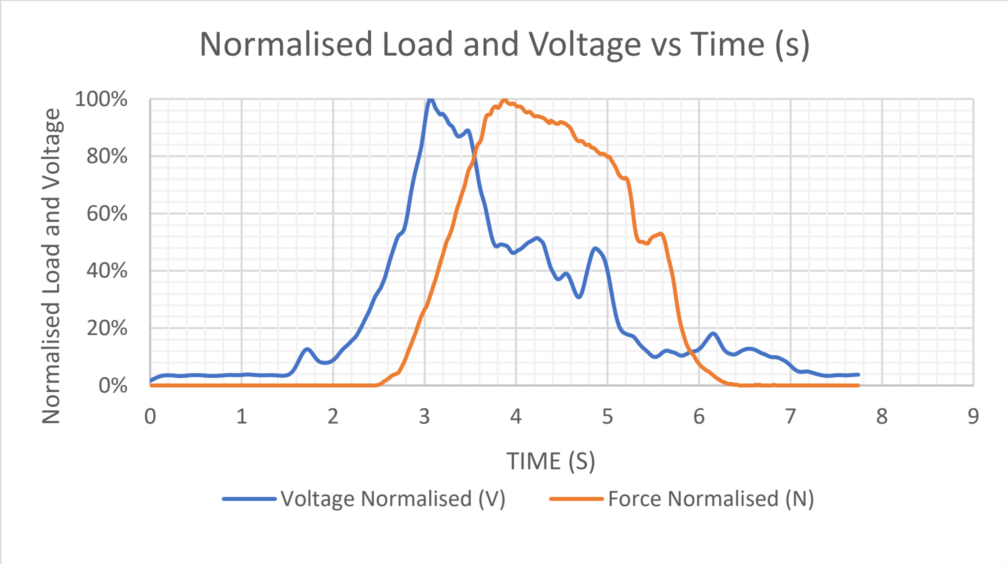



*Figure 22 - Normalised Load (N) and Voltage (V) vs Time (s)*

**Figure 22** depicts a plot of normalised voltage and force with respect to time. It can be noted that the voltage and the load ramp up are out of phase. This delay could due to the stimulus detected by the sEMG unit and the response in the form of the actual force output or due to the stimulus and the response of both electrochemical and mechanical processes. Following the synaptic transmission from the motor unit, a short while is required for ions to experience change in the action potential as well as the Excitation-Contraction coupling of released calcium ions, to shorten the sarcomeres and induce muscular contraction. [11]

Aside from delays, sEMGs only read signals from muscle fibres closest to the skin surface. The electrical signal might be affected by electrical cross talk from other deeper muscles which may also increase according to the amount of muscle fibres present as well as due to voltages generated by contractions of other accessory muscles close to the main muscle itself. [12]
Correlation of force and voltage data requires compensation of the latency experienced by the signals. Points of interest for both the force and sEMG signals were noted, most importantly where voltage and force data started to increase. Force and Voltage sets were cropped and plotted against a normalised time profile as depicted in **Figure 23.**

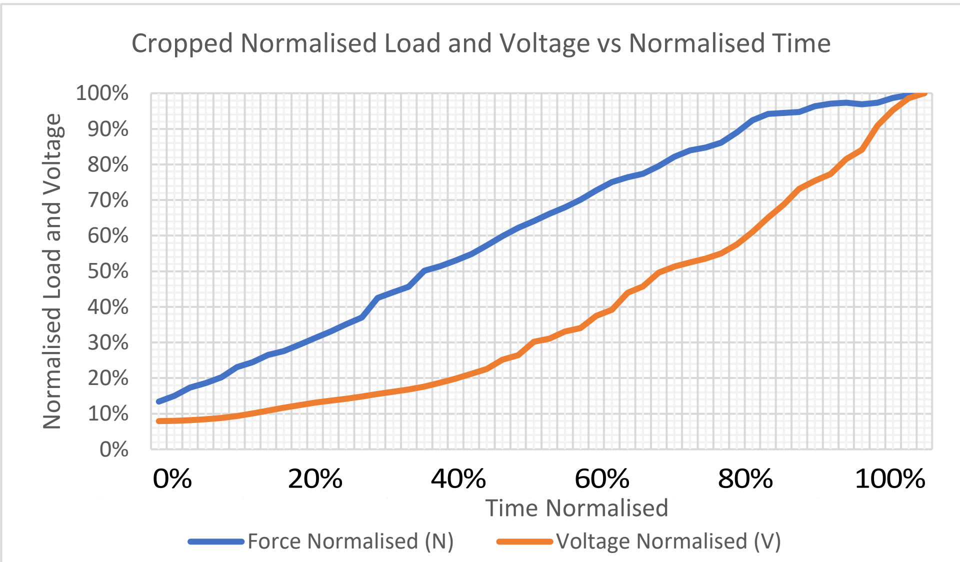



*Figure 23 - Cropped Normalised Load and Voltage vs Normalised Time*

A graph of the force against the sEMG voltages was plotted shown in **Figure 24**Figure 23. A high order polynomial could have been plotted for a best line through the points for better accuracy, but the degree of error would increase drastically. Thus, a zero-intercept, 3rd order polynomial trendline was fitted to the profile such that its equation provided a direct correlation between the force and the voltage of the MVC of the Vastus Medialis.

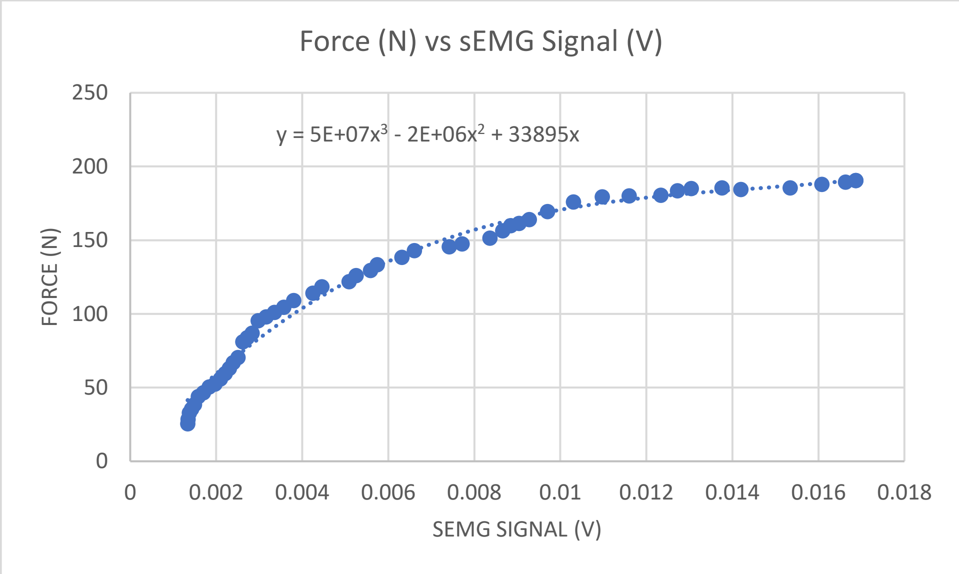



*Figure 24 - Force (N) vs sEMG Signal (V)*

The polynomial obtained was then used to generate the force-voltage profiles for the sEMG data recorded for ten different muscles during the gait experiment. The previously used data which was cropped and imported in the MOKKA Software was imported to Microsoft Excel. To prepare the sEMG data for filtering, the time was normalised and the sEMG signal was rectified again as was done for the MVC of the Vastus Medialis. Again, the Butterworth filter was used on the sEMG data choosing a single cut-off frequency of 12 Hz to maintain the general shape of the curve without causing irregularities or loss of detail. The polynomial was then used to calculate the force from the filtered sEMG value by substituting the filtered data into the equation. The calculated force was used to obtain a profile of the force variation correlated to the sEMG data with respect to the gait cycle % for all muscles recorded during the gait.

### Vastus Medialis

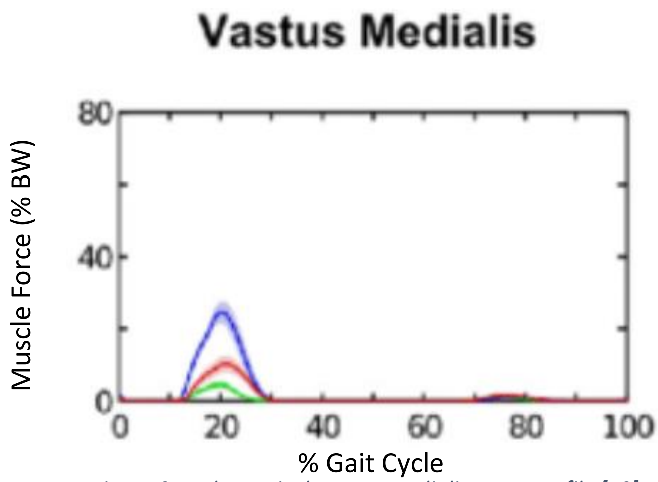



*Figure 25 - Theoretical Vastus Medialis Force Profile [13]*

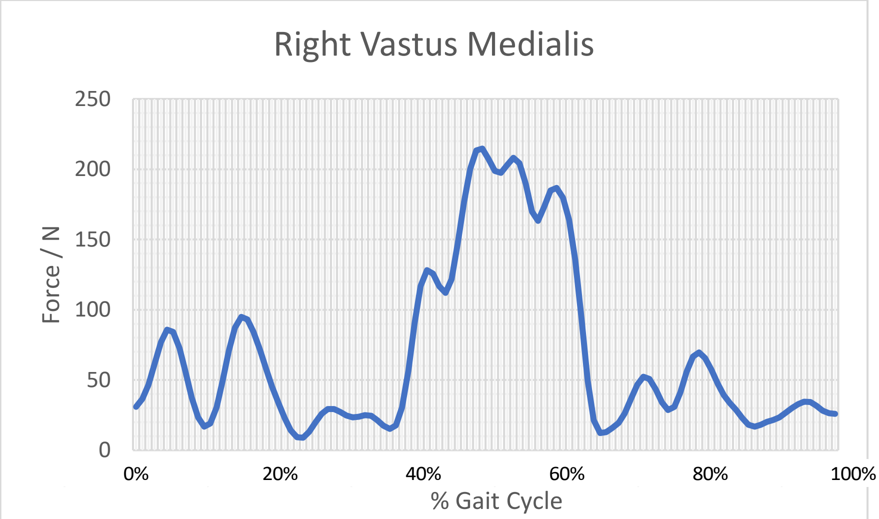



*Figure 26 - Right Vastus Medialis Force Profile*

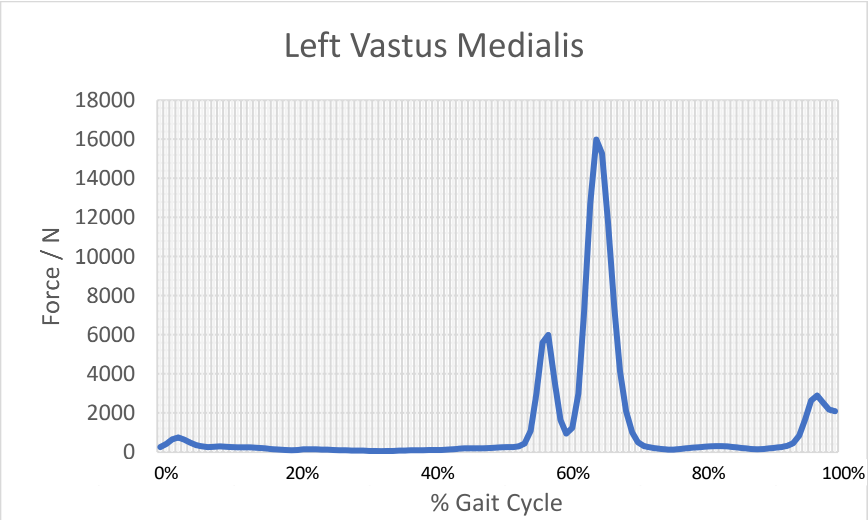



*Figure 27 - Left Vastus Medialis Force Profile*

On comparing the experimental graphs for the Right and Left Vastus Medialis graphs in **Figure 26** and **Figure 27** respectively, to the theoretical graph from **Figure 25**, one can note a considerable difference. Numerous assumptions were made during the force profiles calculations which led to inaccuracies in the resulting data, with the most significant being that the force-voltage relationship from the MVC of the Vastus Medialis is universal to all muscles. As a result, it is assumed that all muscles are equal in size and also have the same force-voltage characteristics. Undoubtedly, superficial and surrounding muscular cross-talk influenced the derived sEMG readings. Since the trendline equation for the sEMG voltage vs force relationship is a 3rd order polynomial, the error would be highly magnified, leading to possible exaggerated force peaks. A case in point for this error is depicted in **Figure 27** where the maximum peak is approximately 16000N. [12] The graph for the right Vastus Medialis (**Figure 26**) is erroneous since the quadriceps would supposedly be contracting eccentrically and concentrically during pre-swing between 50% and 60% of the gait cycle as opposed to the graph where a phase shift is clear. On the other hand, even though the force values are exaggerated, the shape of the graph conforms to the theoretical but with a lagging phase shift of half a cycle of the left referenced leg. Moreover, the minor peaks may be a consequence of cross-talk and minor contractions of the Rectus Femoris since the Vastus Medialis is not the major agonist during walking. [7]

Gastrocnemius Lateralis

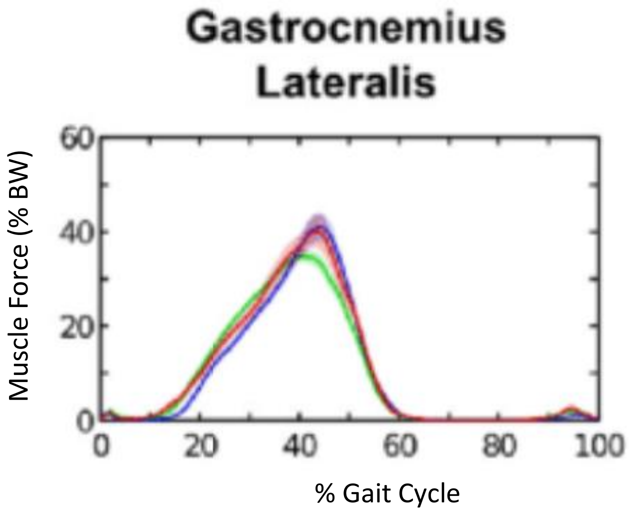



*Figure 28 - Theoretical Gastrocnemius Lateralis Force Profile [13]*

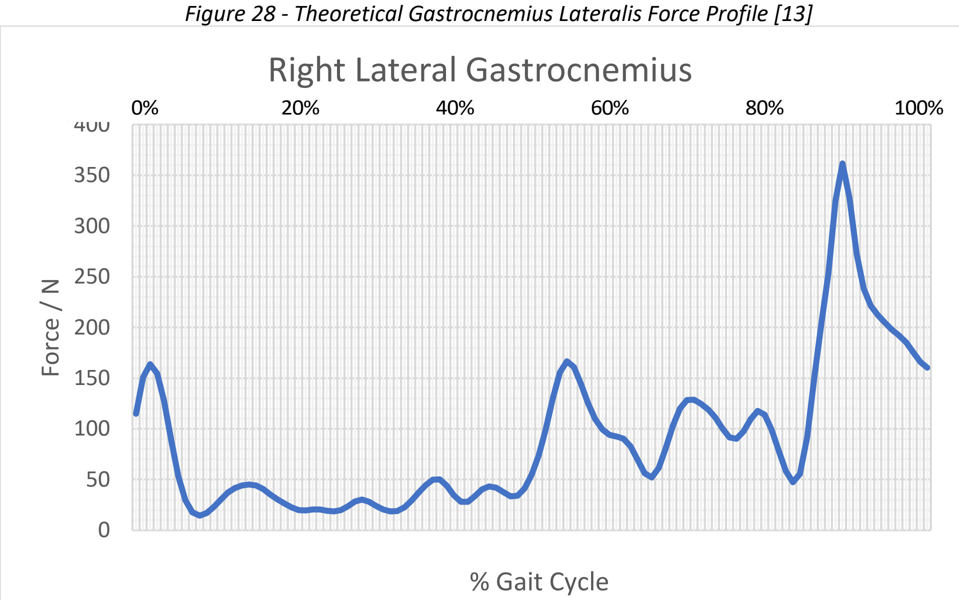



*Figure 29 - Right Lateral Gastrocnemius Force Profile*

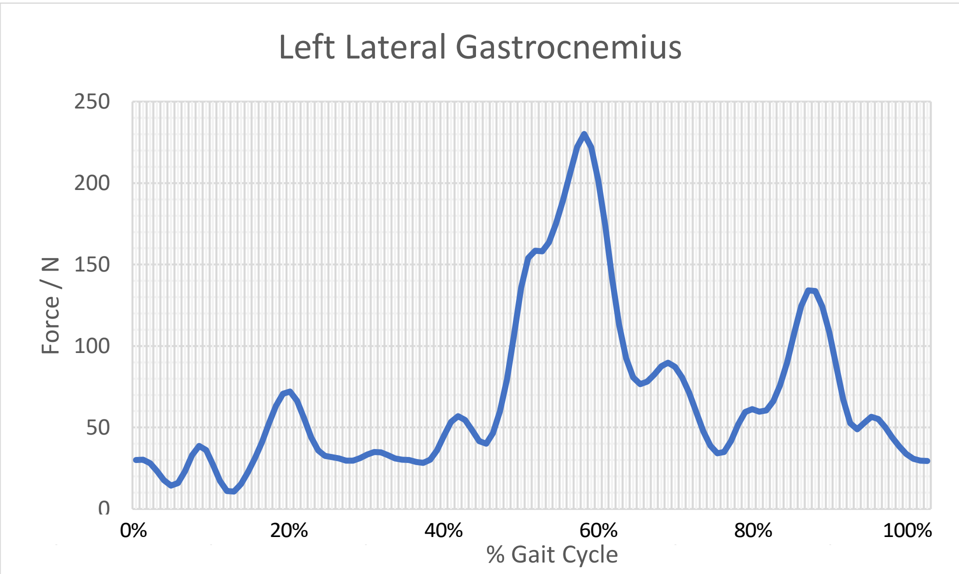



*Figure 30 - Left Lateral Gastrocnemius Force Profile*

The Right Lateral Gastrocnemius depicted in **Figure 29** does not appear to conform with the theoretical force profile in **Figure 28**. The peak seems to trail by half a cycle in comparison to the theoretical force profile. The Left Lateral Gastrocnemius depicted in **Figure 30** gives a closer representation of the theoretical graph as the peak occurs in the mid portion of the gait cycle as in the theoretical. A quite noticeable difference is the lack of smoothness in the experimental graph as a consequence of cross-talk from other surrounding muscles which affects the EMG signal as well as inadequate filtering of the data. The Gastrocnemius Lateralis is used to allow plantarflexion in the ankles as well as to promote flexion in the knee joints. With respect to the gait cycle, the gastrocnemius is mainly used during toe off which occurs at approximately 90% and 60% of the gait cycle for the Right and Left Lateral Gastrocnemius respectively as depicted by **Figure 30**. The force generated in the experimental graphs is comparable to the theoretical force, showing that the force-voltage correlation from the Vastus Medialis is quite identical. A considerable difference in force generated can be noted between the left and right Lateral Gastrocnemius depicting a muscular imbalance or dysfunction.

## Anterior Tibialis

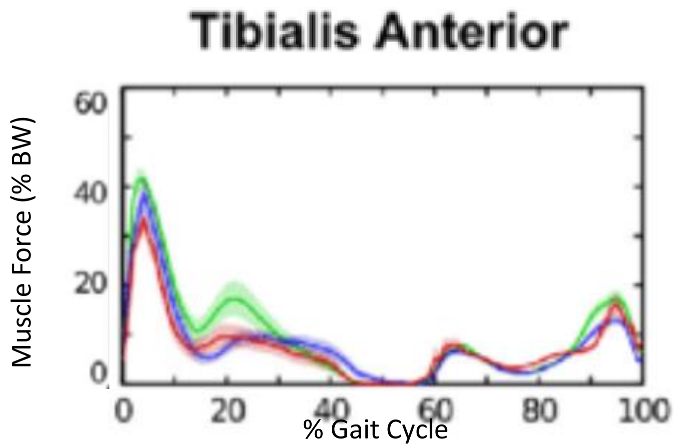



*Figure 31 - Theoretical Anterior Tibialis Force Profile [13]*

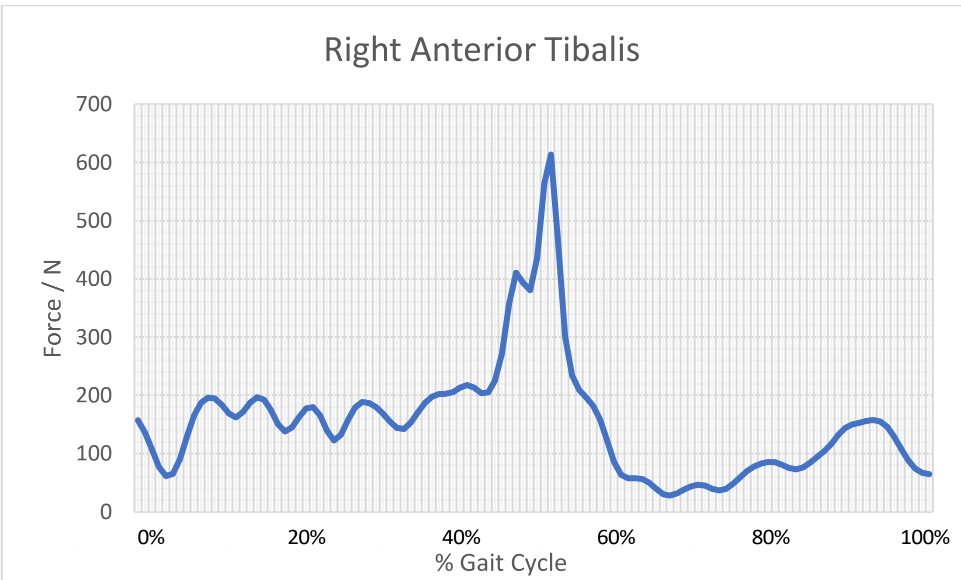



*Figure 32 - Right Anterior Tibialis Force Profile*

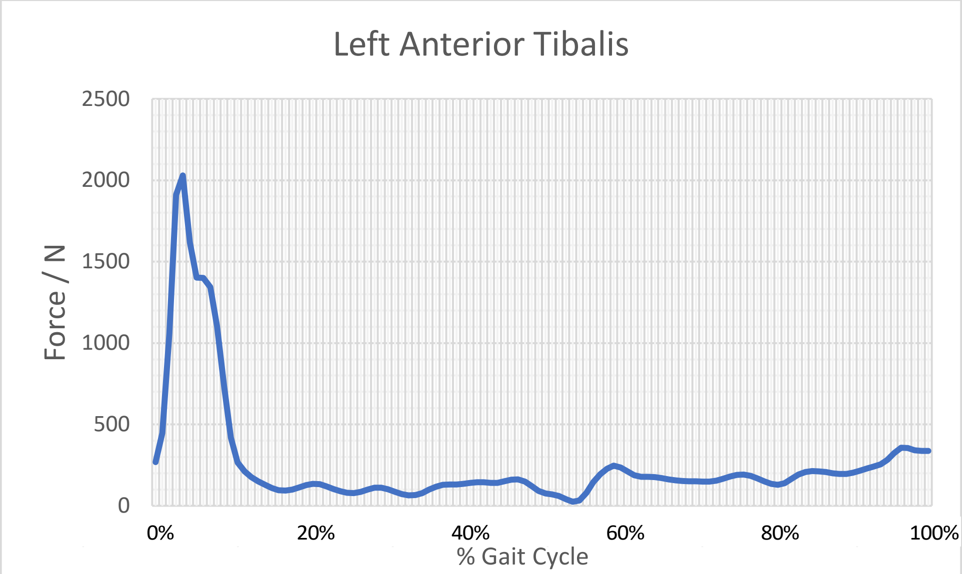


*Figure 33 - Left Anterior Tibialis Force Profile*

A significantly large surge in the sEMG signal was noted for both the left and right Anterior Tibialis muscles depicted by **Figure 32** and **Figure 33** respectively possibly due to distortions in the readings. However, the shape of the graphs is comparatively similar to the theoretical graph depicted in **Figure 31**. The initial peak of the Left Anterior Tibialis graph corresponds to the force generated by the eccentric contraction to promote ankle dorsiflexion, which is also present in the Right Anterior Tibialis lagging by half a cycle. The smaller peaks are due to the concentric contraction of the Anterior Tibialis to resist the external moments. These peaks are relatively larger to the forces depicted in the theoretical graph, possibly due to the test subject's constant cocked ankle during his gait. Between 0% to 10% and 40% to 60% of the Right and Left Anterior Tibialis Force profiles, a drop in the muscular force occurs as a result of activation of the triceps Surae during the terminal stance and the pre-swing. This drop in force occurs due to the simultaneous activation of the Triceps Surae whilst the body is shifted on one during forward propulsion. [7]

## Gastrocnemius Medialis

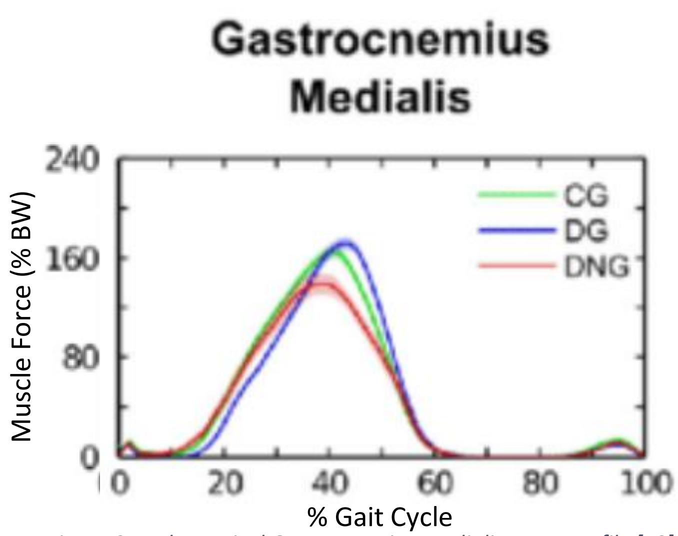


*Figure 34 - Theoretical Gastrocnemius Medialis Force Profile [13]*

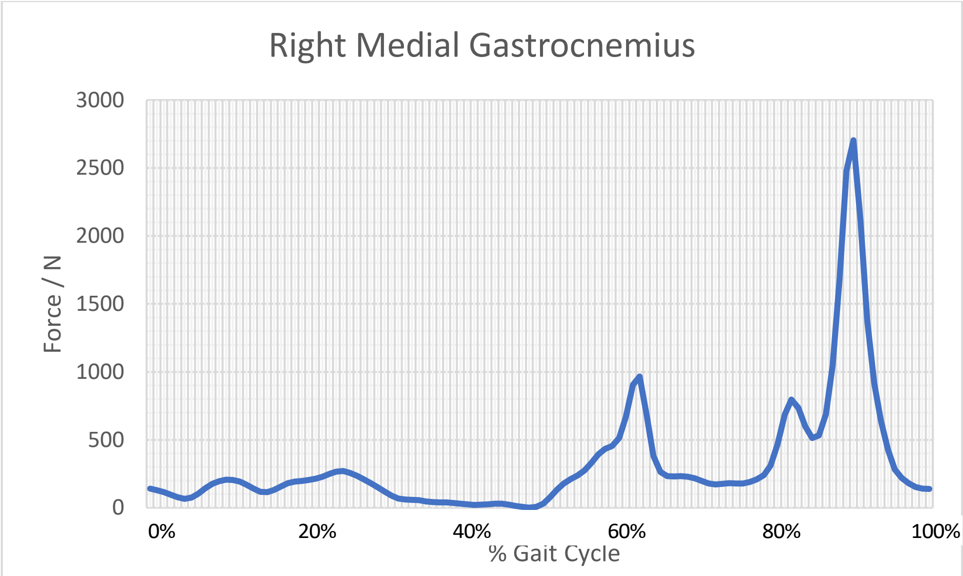



*Figure 35 - Right Medial Gastrocnemius Force Profile*

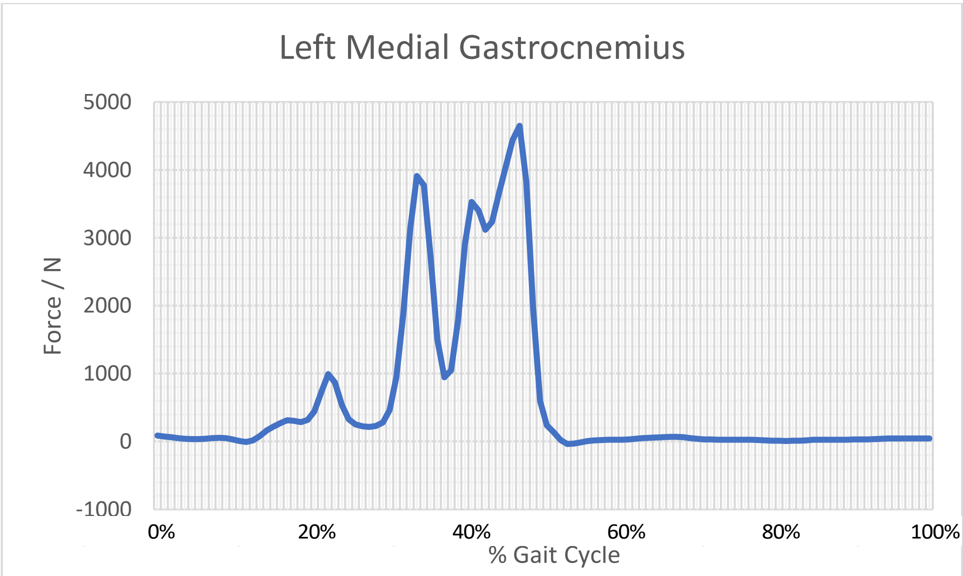



*Figure 36 - Left Medial Gastrocnemius Force Profile*

Both **Figure 35** and **Figure 36** of the Right and Left Medial Gastrocnemius seem to have similar characteristics to the Theoretical graph in **Figure 34**. Even though a massive surge in Force occurred in both graphs, the Left Medial Gastrocnemius tended to peak between 30% to 60% of the gait cycle as opposed to the 20% to 60% of the Theoretical. During this time of the gait, the test subject underwent Mid Stance to Pre-Swing where the Medial Gastrocnemius was used to propel the body forward as a result of the contraction via dorsiflexion after the initial heel strike, increasing plantarflexion in the ankle to propel the body forward until toe-off. The erroneous for magnitudes could be attributed to the previously mentioned assumptions as well as sEMG signal contamination which would lead to sudden oscillations in the voltage and hence, the force.
Due to the test subject's tendency of plantarflexion rather than attaining a neutral ankle, coupled with the shorter strides, the gastrocnemius contraction might have been delayed and performed more abruptly in the terminal stance phase. [8] [16] Moreover, muscular crosstalk attributed to unstable oscillations throughout the cycle or possibly due to loss of stability or inappropriate data filtering when converting the sEMG voltage to force.

Bicep Femoris Short Head

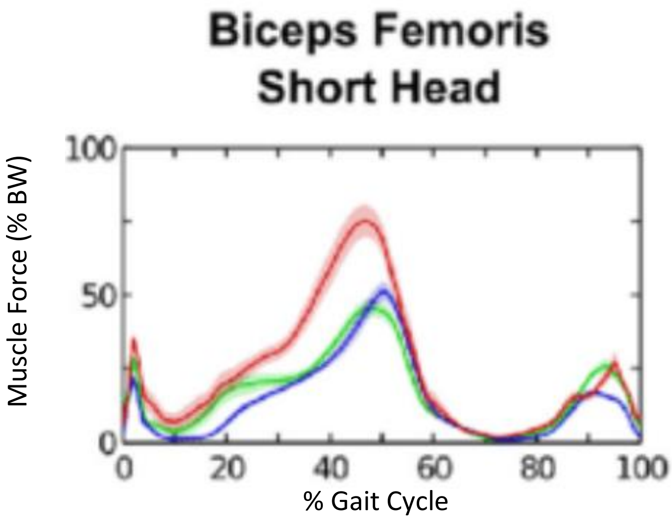



*Figure 37 - Theoretical Bicep Femoris Short Head Force Profile [13]*

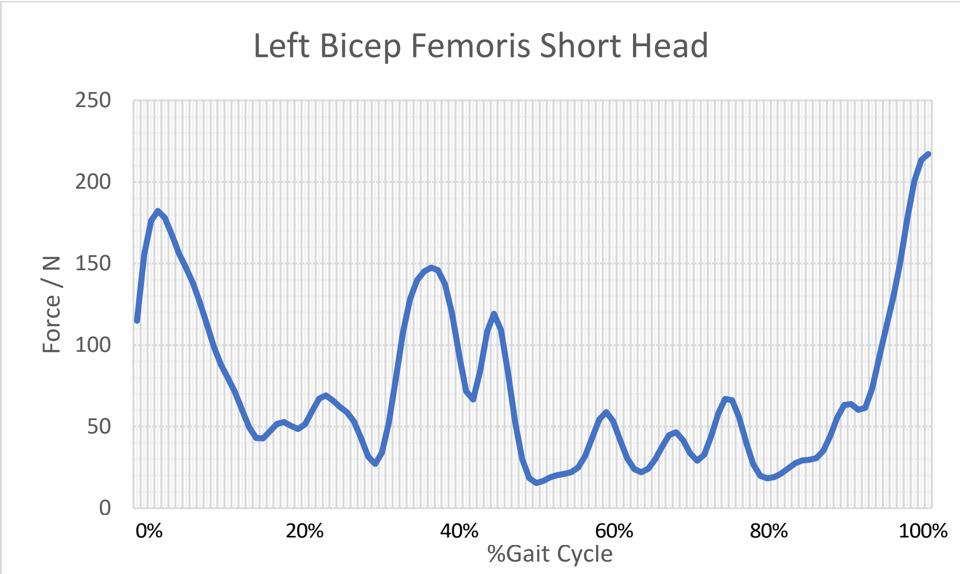



*Figure 38 - Left Bicep Femoris Short Head Force Profile*

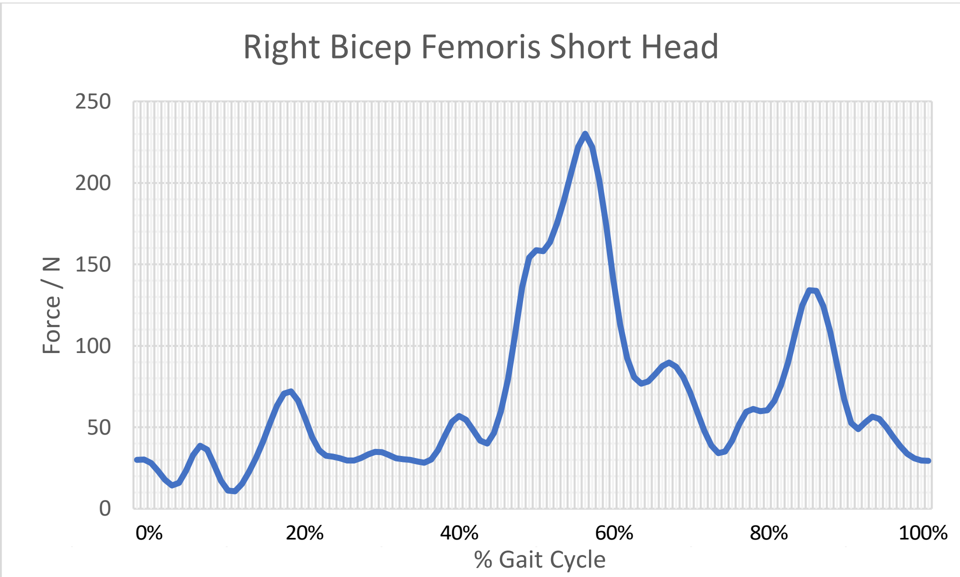



*Figure 39 - Right Bicep Femoris Short Head Force Profile*

Both **Figure 38** and **Figure 39** of the Right and Bicep Femoris Short Heads seem to have similar characteristics to the Theoretical graph in **Figure 37**. The force profiles. It can be seen that the largest peaks for the Biceps Femoris graphs occur during the loading response and in the terminal swing phase between approximately 0% to 15% for the Left Bicep Femoris and approximately half a cycle later for the right Biceps Femoris between 50% to 70%. The peaks make sense given that the hamstring contracts to prevent hyperextension of the knees. The smaller peaks occur mainly due to cross-talk as well as the shorter stride. [7]

Comparative values for the magnitudes of both sides can be seen from the graphs depicting equal work done by both sides. However, the forces depicted by the theoretical biceps femoris short head graphs in **Figure 37** are considerably higher than the forces represented by the peaks in the experimental graphs. Given that the theoretical graph provides the peak at approximately 60% bodyweight, and the participant weighed 82.97 kg at the time of the experiment, a peak of approximately 225N for both graphs showed non-conformance to the theoretical estimate of 500N. This erroneousness is primarily due to the force-voltage assumptions taken during the experiment and the analysis of the results. [7]

Bibliography

[1] Musculoskeletalkey.com. (2018). [online] Available at: https://musculoskeletalkey.com/wp-content/uploads/2016/12/B9780702042652000021_f02-01-9780702042652.jpg [Accessed 20 Dec. 2018].

[2] ArtBot. (2018). A study of the human gait cycle in order to design a biped robot. [online] Available at: https://sme-chinoises-euronext.typepad.fr/artbot/2014/09/a-study-of-walking-in-order-to-design-a-biped-robot.html [Accessed 20 Dec. 2018].

[3] Streifeneder.com. (2018). The Human Gait Cycle. [online] Available at: https://www.streifeneder.com/downloads/o.p./400w43_e_poster_gangphasen_druck.pdf [Accessed 9 Dec. 2018].

[4] Musculoskeletalkey.com. (2018). [online] Available at: https://musculoskeletalkey.com/wp-content/uploads/2016/12/B9780702042652000021_f02-09-9780702042652.jpg [Accessed 20 Dec. 2018].

[5] Zflomotion.com. (2018). [online] Available at: https://www.zflomotion.com/hs-fs/hub/167460/file-2574763968-jpg/grfs-graph.jpg [Accessed 20 Dec. 2018].

[6] P. Konrad, The ABC of EMG. Noraxon USA, 2006, pp. 5-15 [Accessed 20 Dec. 2018].

[7] M. Whittle, Gait analysis. Edinburgh [u.a.]: Butterworth-Heinemann, 2007, pp. Chapter 2

[8] Workplace, N. (2018). *Biomechanics*. [online] Ncbi.nlm.nih.gov. Available at: https://www.ncbi.nlm.nih.gov/books/NBK222434/ [Accessed 17 Dec. 2018].

[9] Phys.org. (2018). *New research finds that world-class sprinters attack the ground to maximize impact forces and speed*. [online] Available at: https://phys.org/news/2014-08-world-class-sprinters-ground-maximize-impact.html [Accessed 18 Dec. 2018].

[10] Jason Cholewa - Assistant Professor of Exercise Science at Coastal Carolina University. (2018). *Optimal Sprinting Performance: A Foot Fetish*. [online] Available at: https://jasoncholewa.com/2012/11/02/optimal-sprinting-performance-a-foot-fetish/ [Accessed 18 Dec. 2018].

[11] H. Begovic, G. Zhou, T. Li, Y. Wang and Y. Zheng, "Detection of the electromechanical delay and its components during voluntary isometric contraction of the quadriceps femoris muscle", Frontiers in Physiology, vol. 5, 2014. [Accessed 19 Dec. 2018].

[12] A. Hof, "The relationship between electromyogram and muscle force", Sportverletzung · Sportschaden, vol. 11, no. 03, pp. 79-86, 1997.

[13] Ackermann, M. (2018). [online] Researchgate.net. Available at: https://www.researchgate.net/profile/Marko_Ackermann/publication/320959866/figure/fig2/AS:614065806262272@1523416105153/Mean-1-standard-error-of-the-force-time-series-for-the-knee-flexors-and-hip-extensor.png [Accessed 19 Dec. 2018].

[14] Cdn-content.qualisys.com. (2018). [online] Available at: https://cdn-content.qualisys.com/2014/01/gait-setup-3d-1200x675.jpg [Accessed 20 Dec. 2018].